

When Inequality Meets Finance: Household Portfolios and the Corporate Bond Boom^{*}

Maria Teresa Sarmiento-Villena[†]

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Abstract

How does rising income inequality reshape financial markets? This paper demonstrates that shifts in the income distribution can influence corporate debt structure by altering household saving behavior. When labor income risk rises and increases income inequality, households accumulate more liquid wealth, held in deposits and corporate bonds, as a precautionary motive. However, portfolio adjustment frictions make the composition of this wealth differ across the income distribution. In a quantitative model with heterogeneous households and firms facing financial frictions due to moral hazard, this reallocation towards bonds reduces the liquidity premium, the extra return paid by bonds relative to deposits, and pushes firms toward greater reliance on bond financing. The model explains 59.5% of the observed increase in bond reliance and 57.9% of the decline in the liquidity premium in U.S. data since 1989. Decomposing the mechanisms, higher income risk directly explains 37.5% of the change in corporate bond reliance, while general equilibrium price effects explain the remaining 62.5%. These findings reveal a new transmission channel from inequality to credit markets, operating through the composition, rather than just the level, of household savings, with broad redistributive and financial-structural implications.

JEL Codes: D31, E21, E44, G32

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[†]University of Texas at Austin - Department of Economics (email: mtsarmiento@utexas.edu).

1 Introduction

A growing body of research has documented the persistent rise in income inequality in the United States (U.S.) and explored its macroeconomic implications, from its impact on natural interest rates to aggregate output.¹ At the same time, the corporate sector has undergone a striking transformation. Firms have increasingly shifted away from traditional bank loans toward bond financing.² Yet one important question remains unexplored: how has the long-run rise in income inequality shaped this structural shift toward corporate bonds over traditional bank lending?

The existing literature has primarily examined the empirical relationship between income and consumption inequality, as it sheds light on households' ability to buffer income shocks and thus provides a rationale for government intervention.³ However, little is known about the pass-through from rising income inequality to the interest rates on deposits and corporate bonds, the key financial returns shaping households' liquid wealth.⁴ Understanding the mechanisms behind this pass-through is crucial for both economists and policymakers, as it influences households' portfolio decisions and, ultimately, their ability to smooth consumption and sustain welfare.

This paper contributes by proposing a novel mechanism that links income inequality, corporate debt structure, and real interest rates. By emphasizing the interaction between household portfolio choices and firm-level financing decisions, it highlights an underexplored channel through which inequality affects corporate leverage, the liquidity premium, and the distribution of liquid wealth (deposits and bond savings) across households. The core contribution is that higher income risk shifts the composition of household liquid wealth toward bonds, compressing the liquidity premium, lowering bond yields, and ultimately inducing firms to rely more heavily on bond financing because it becomes relatively cheaper than bank credit.

The paper begins by motivating the mechanism with three stylized facts from U.S. data. First, there is a positive comovement between income inequality and the corporate debt structure. Second, the deposit shares, measured as the share of deposits in total liquid wealth, have evolved heterogeneously across the income distribution. This pattern highlights two salient features. On the one hand, households in the lower part of the income distribution tend to hold a greater share of deposits relative to those at the top. On the other hand, this suggests the presence of frictions or

¹Piketty et al. (2018); Kuhn et al. (2020); Kopczuk et al. (2010); Mian et al. (2021); Auclert et al. (2018)

²Buchak et al. (2024); Crouzet (2021)

³Attanasio et al. (2012); Aguiar and Bils (2015); Heathcote et al. (2010a); Meyer and Sullivan (2023)

⁴According to the Distributional Financial Accounts for 1989-2019, households' liquid wealth, comprising deposits and corporate bond holdings, accounts on average for 26% to 50% of total wealth excluding housing.

inertia in portfolio rebalancing, even among assets typically classified as liquid.⁵ Third, the spread between the return on corporate bonds and short-term commercial paper, a proxy for the liquidity premium in corporate credit markets, has declined over time. This downward trend mirrors the decreasing cost differential for firms choosing between bond and bank financing

Based on these facts, I then develop a model with heterogeneous households that face portfolio adjustment frictions and liquidity needs, allocating wealth differently depending on their labor income and initial liquid wealth. To capture the rigidity of portfolio rebalancing, the model assumes that households face stochastic adjustment costs, drawn from a common distribution, when changing their bond holdings. Consequently, households optimally weigh the marginal benefits against the costs of adjustment, resulting in infrequent and lumpy rebalancing of their bond portfolios.⁶ Thereby, in the model, although both deposits and bonds mature in one period, deposits are modeled as more liquid because households can access them without frictions at any time with perfect flexibility. In contrast, bond holdings can only be adjusted with some probability, creating an endogenous friction. The liquidity premium in equilibrium thus reflects the value households place on the ability to reallocate funds flexibly, rather than differences in maturity.

Moreover, firms are subject to idiosyncratic shocks that affect both their production and the value of assets pledged as collateral. These shocks generate repayment risk and exacerbate moral hazard, thereby creating financial frictions in the contracting relationship between firms and intermediaries. Financial intermediaries (traditional banks and bond markets) channel household savings into firm credit, but they have notable distinctions. Banks possess monitoring technologies that mitigate informational asymmetries, allowing them to pool risk across firms and ensure repayment to households. Firms, in turn, endogenously adjust the mix of bank loans and bond issuance in response to shifts in the relative cost of these credit sources. This feedback loop generates a novel transmission channel from inequality to real interest rates.

The baseline estimation of the model is disciplined by a set of targeted moments that capture both household portfolio patterns and firm financing behavior in the U.S. economy around 1990. On the household side, the discount factor, portfolio frictions, and borrowing penalty parameters are estimated to match the aggregate and cross-sectional deposit shares across the income distribution, as well as the share of borrowers. On the firm and banking side, parameters governing idiosyncratic and exit shocks, production technology, monitoring intensity, and collateral constraints are calibrated to match external funding relative to net worth, the use of bank loans and collateral,

⁵Bayer et al. (2019); Kaplan et al. (2018) treat both deposits and corporate bonds as liquid assets.

⁶These adjustment costs can be viewed as representations of risk premia, behavioral biases or informational frictions that discourage households from rebalancing when they misperceive the risks associated with bonds.

bond returns, and the liquidity premium. Together, these moments ensure that the stationary equilibrium provides a realistic benchmark for analyzing the effects of rising income inequality.

In my main exercise, I increase the dispersion of idiosyncratic labor productivity shocks to generate the observed rise in income inequality in a new stationary equilibrium. After increasing labor income risk to reflect higher income inequality, the model delivers a strong quantitative fit to key macro-financial patterns observed in the U.S. between 1989 and 2019. It generates the link between rising income inequality, and firms' increasing reliance on bond financing, accounting for 59.5% of the observed change in firms' financing. The model also captures the heterogeneous reallocation of household savings across the income distribution, including the decline in deposit intensity among top households and the relative shift toward deposit holdings at the bottom. Moreover, it partially matches the empirical reduction in the liquidity premium, accounting for 57.9% of the observed drop. The model also offers a structural explanation for changes in household financial intermediation and the evolution of the corporate debt structure. These results underscore the connection between household portfolios and aggregate credit market outcomes. Even under simplified assumptions, they provide a foundation for analyzing structural transformations in financial markets.

Decomposing the core mechanism reveals that about 37.5% of the total change is directly driven by higher income risk, while the remaining 62.5% stems from general equilibrium feedback through prices. In the baseline economy, before inequality rises, firms rely on bond financing for 32.0% of their debt. When income risk increases but prices are held fixed, bond reliance rises modestly to 33.8%. Allowing prices to adjust further amplifies this effect, increasing bond reliance to 36.8%.

Another important lesson from the model is that rising income inequality can generate redistributive effects on household portfolios, resulting in a compression of liquid wealth inequality. While standard intuition suggests that greater labor income dispersion should amplify disparities in financial asset holdings, the model instead predicts a decline in the Gini coefficient of liquid wealth, from 0.42 to 0.40. This outcome stems from a decrease in the liquidity premium, which raises the marginal value of bond holdings for households in the upper-middle of the distribution, enabling them to rebalance their portfolios toward bonds. By contrast, households at the bottom remain concentrated in deposits, while those at the top, already heavily invested in bonds, face a low marginal value of additional bond holdings and increase their positions only marginally. These offsetting responses yield a more balanced distribution of liquid wealth and reveal a new channel through which inequality shapes financial markets by narrowing gaps in liquid wealth and compressing liquidity premia.

Moreover, the comparison between partial and general equilibrium reveals how market feedback reshapes household portfolios. In partial equilibrium, higher income risk primarily reduces deposit shares among top-income households while slightly increasing them among low-income earners. Once general equilibrium price changes are introduced, however, middle-income households become the main drivers of reallocation, shifting sharply toward bonds as their marginal value of holding them rises.

This reversal shows that inequality’s effects on portfolio composition are amplified through general equilibrium forces that reshape the distribution of liquid wealth. Liquid wealth inequality widens between upper-middle and lower-income households, as the former accumulate bonds more rapidly, but narrows between the top 1% and the rest, reflecting their limited gains from falling bond returns. Consumption inequality takes a hump-shaped form: lower and middle-income households experience modest gains, upper-middle households benefit the most, and the top 1% face sharp declines due to their heavy exposure to falling bond yields. Income inequality mirrors this pattern, rising between the bottom and upper-middle but narrowing between the top 1% and the rest, as shifting financial returns redistribute resources away from the very top. Overall, the surge in inequality reduces welfare for all, though unevenly: on average, households would need a 4.04% permanent increase in consumption to be as well off as before, with deposit-poor and bond-rich households suffering the largest losses.

Contribution to the literature: This paper contributes to three main strands of the literature. First, this paper complements and extends the growing literature on income and wealth inequality (Mian et al., 2020; Auclert et al., 2021; Kuhn et al., 2020; Kopczuk et al., 2010; Smith et al., 2023; Gomez, 2025; Fernandez-Villaverde et al., 2024; Cao and Luo, 2017; Fisher et al., 2020) by linking distributional dynamics to the structure of firm financing. It also complements empirical evidence that shows that consumption inequality has closely mirrored the rise in income inequality in the U.S. over recent decades (Aguiar and Bils, 2015; Attanasio et al., 2012). Moreover, recent studies have emphasized the redistributive effects of macroeconomic shocks. For instance, Coibion et al. (2017) shows that contractionary monetary policy disproportionately affects low-income households, thereby exacerbating income and consumption inequality. Similarly, Heathcote et al. (2010a,b) document a persistent rise in income inequality and labor income risk since the 1980s. These findings underscore the importance of understanding how rising income risk shapes household saving behavior and its macroeconomic consequences, an interaction that is central to the mechanism explored in this paper.

While these studies document and explain the evolution of inequality and its implications for aggregate demand, savings, and wealth distribution on the top end, this paper focuses on how

shifts in the joint distribution of household income and liquid asset holdings affect the composition of corporate debt, specifically, the relative reliance on bank loans versus bond financing. By embedding heterogeneous households and firms in a general equilibrium framework, the model uncovers a novel transmission channel. Rising inequality reallocates savings toward upper-middle-wealth households with stronger preferences for less liquid, higher-yield assets (e.g., corporate bonds), thereby reshaping the supply of intermediary capital and altering firm debt structure. This perspective adds a new dimension to the inequality debate by showing how financial structure, not just macroeconomic aggregates, responds endogenously to changes in the income distribution.

Second, it builds on heterogeneous-agent models with two-asset structures. While recent research (Doerr et al., 2024; Bayer et al., 2019; Kaplan et al., 2014, 2018; Auclert et al., 2024) has examined how heterogeneity in household saving behavior shapes macroeconomic outcomes, such as employment, investment, and consumption, this paper focuses on a complementary but previously underexplored mechanism. I develop a general equilibrium model in which households, differentiated by income and initial liquid wealth face portfolio adjustment frictions and choose how to allocate their savings across assets with different liquidity properties.⁷ These heterogeneous frictions shape the aggregate composition of household savings, particularly the balance between deposits and corporate bonds, which in turn affects firms’ relative reliance on bank loans versus bond financing. Unlike much of the existing literature, which abstracts from within-firm debt composition, this paper explicitly models how household heterogeneity gives rise to endogenous corporate debt structure decisions.⁸ This interaction introduces a novel mechanism through which income inequality affects the liquidity premium, the cost of capital, and ultimately the equilibrium real interest rate.

Third, it contributes to the literature on financial frictions and financial heterogeneity. Prior studies (Holmstrom and Tirole, 1997; Kiyotaki and Moore, 1997; Rampini and Viswanathan, 2019; Villacorta, 2018; Fontaine and Garcia, 2012; Gromb and Vayanos, 2018; He and Krishnamurthy, 2013) emphasize the role of intermediary capital in shaping firm leverage and financial dynamics. While many of these works focus on the joint evolution of firm and bank net worth or attribute financial disruptions to common shocks affecting intermediaries’ or speculators’ wealth, thereby constraining their capacity to provide liquidity or engage in arbitrage, this paper abstracts from bank net worth and instead emphasizes long-run general equilibrium effects arising from shifts in

⁷Kaplan et al. (2018) and Bayer et al. (2019) explore household portfolio choices between illiquid assets (e.g., housing or equity) and liquid assets (e.g., deposits or bonds). In contrast, this paper focuses on the macroeconomic implications of changes in the composition of liquid household assets, specifically, deposits and corporate bonds.

⁸Doerr et al. (2024) also studies firm-level debt composition, but focuses on differences across firm types (e.g., public vs. private), whereas this paper examines variation within firms driven by household-side heterogeneity.

the composition of household savings.

Furthermore, it contributes to the literature on financial accelerator proposed by [Bernanke et al. \(1998\)](#) that emphasizes the importance of the external finance premium, which is the difference between the cost of external funds (borrowing) and the opportunity cost of internal funds (retained earnings). In this paper, the comparison between the respective sources of external financing (bank loans and bonds) and the return on capital is also presented, and it is key for firms to determine their optimal leverage and debt structure. However, this external finance premium depends on the endogenous decision of households to balance their portfolios of deposits and bonds.

The remainder of the paper is organized as follows. Section 2 presents the motivating empirical evidence. Section 3 develops the quantitative model and derives closed-form solutions for key credit variables. Section 4 discusses the numerical implementation and describes the model estimation, Section 5 analyzes the quantitative results, and Section 6 provides the conclusion. The appendices offer additional technical details and supporting material for the sections throughout the paper.

2 Motivating evidence

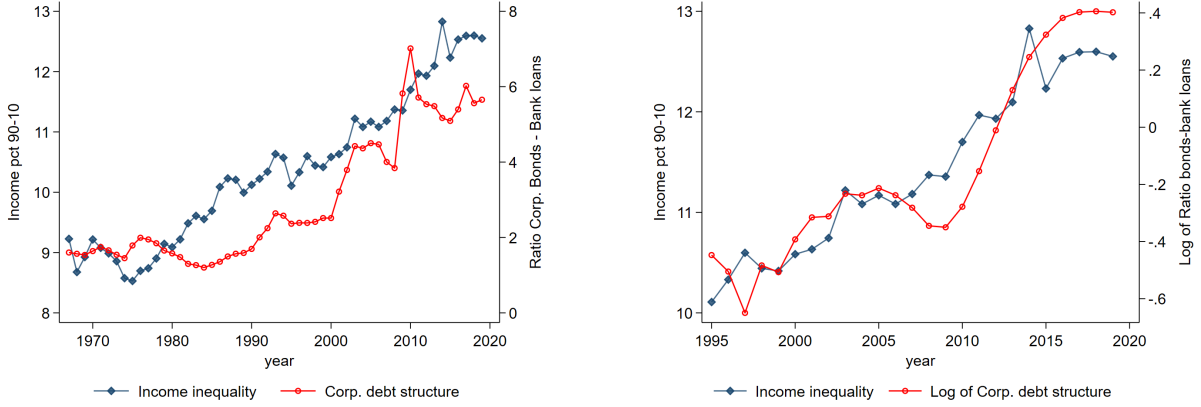
The quantitative model is motivated by three empirical patterns in the U.S. economy. First, there is a positive co-movement between income inequality and the corporate debt structure. Second, the heterogeneous composition of liquid wealth across households, particularly in the allocation between deposits and bonds. Third, the downward trend in the corporate liquidity premium.

2.1 Co-movement between inequality and corporate debt structure

Over the past five decades, income inequality in the U.S. has risen steadily. During this same period, firms have gradually shifted away from traditional bank lending and increasingly relied on bond markets for external financing, a trend reflected in the rising ratio of corporate bonds to bank loans, as shown in Figure 1. Data on income inequality are drawn from the Current Population Survey (CPS), while information on corporate debt structure comes from the Z.1 Financial Accounts at the aggregate level and from Capital IQ at the firm level.⁹

⁹Using the transaction descriptions in Capital IQ, I classify as bank loans those labeled as "Bank loans", "Term loan (facility)", "Bank overdraft (facility)", "Revolving credit (facility)", "Letter of Credit Facility" and "Letter of Credit Outstanding" and "Commercial paper (facility)". I classify as bonds those labeled as "Bonds and notes", "Debentures", and "Notes payable (facility)".

Figure 1: Income inequality and corporate debt structure.



(A) Co-movement at aggregate level

(B) Co-movement at firm level

Notes: Figure A plots the evolution of income inequality (blue line), together with the ratio of corporate bonds to bank loans in the liabilities of non-financial corporations (red line), using aggregate data from the Current Population Survey (CPS) and the Z.1 Financial Accounts. Figure B presents the firm-level counterpart, illustrating the co-movement between income inequality and the 3-year moving average of the log-ratio of corporate bonds to bank loans, constructed from the CPS and Capital IQ databases. In both cases, income inequality is measured as the ratio of the 90th-to-10th percentile of the income distribution, allowing a direct comparison between aggregate and firm-level patterns.

This transformation in corporate debt structure is observable not only in aggregate data but also at the firm level, indicating a broad and persistent change in how firms, particularly larger ones, access credit.¹⁰ Notably, the increasing reliance on bond financing occurred during a period of historically low interest rates, when both bank loans and bonds became cheaper. This suggests that when interest rates are very low, the shift in debt structure reflects differing elasticities of demand across the two debt instruments.

Using U.S. time-series regressions for the period 1967-2021 from the Z.1 Financial Accounts and the Current Population Survey (CPS), I estimate the relationship between various measures of income inequality and the aggregate ratio of corporate bonds to bank loans, controlling for a linear time trend t :

$$y_t = \beta_0 + \beta_1 Ineq_t + \beta_2 t + \epsilon_t \quad (1)$$

where the dependent variable y_t is the ratio of corporate bonds to bank loans for non-financial

¹⁰The rising log-ratio of corporate bonds to bank loans at the firm level reflects a growing reliance on bond financing relative to bank credit.

corporate businesses, and $Ineq_t$ denotes alternative time-varying measures of income inequality, specifically, the income ratios of the 80th to 10th percentile, 90th to 10th percentile, and 95th to 10th percentile.

Table 1: Relation between the ratio of corporate bonds-to-bank loans and income inequality

	(1)	(2)	(3)
	Corp.Bonds-Banks	Corp.Bonds-Banks	Corp.Bonds-Banks
Time-trend	0.0565*** (0.0178)	0.0419* (0.0215)	0.0487** (0.0241)
Disposable income ratio 80:10	0.937** (0.365)		
Disposable income ratio 90:10		0.679*** (0.243)	
Disposable income ratio 95:10			0.381** (0.169)
Observations	55	55	55
R ² adj.	0.807	0.809	0.803

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each specification is estimated separately for the three inequality measures, following the model in Equation 1. The standard errors are adjusted for heteroskedasticity, allowing for valid statistical inference even when the variability of the error terms changes over time.

The results in Table 1 indicate a strong and statistically significant positive association between income inequality and the relative importance of bond financing. Specifically, in column (1), a one-unit increase in the 80th-to-10th percentile income ratio is associated with a 0.937 increase in the corporate bond-to-bank loan ratio, controlling for the linear time trend. The corresponding estimates for the 90th-to-10th and 95th-to-10th ratios are 0.679 and 0.381, respectively. All three specifications capture over 80% of the variation in the dependent variable, as reflected in the adjusted R^2 .

Importantly, this association is not confined to the U.S. Cross-country evidence also reveals a positive relationship between disposable income inequality and the structure of credit markets. Using annual data from the Global Financial Development Database and the World Inequality Database, I estimate pooled OLS regressions to explore this relationship:

$$y_{it} = \beta_0 + \beta_1 Ineq_{it} + \beta_3' X_{it} + u_t + v_i + \epsilon_{it} \quad (2)$$

where the dependent variable y_{it} denotes alternative measures of bond market structure for country i in year t , including the share of domestic non-financial corporate bonds in total domestic bonds outstanding (Corp. bonds %) and the annual size of the public bond market capitalization (Bond Mkt Cap). The variable $Ineq_{it}$ captures different measures of income inequality, such as the S80/S20 ratio (the share of disposable income held by the top 20% relative to the bottom 20%) for the total and working age population, and the Gini coefficient of disposable income (after and before taxes and transfers) for the total and working age population. To control for income levels and the size of the credit market, I include lagged GDP per capita and the lagged ratio of total liabilities to GDP, and u_t and v_i are time and country fixed effects, respectively.

One limitation of the datasets used in Table 2 is the absence of a consistent and well-defined variable capturing the volume of traditional bank loans, which restricts a direct analysis of bond vs. bank financing. Despite this constraint, Table 2 offers strong empirical support for the hypothesis that rising income inequality is associated with a greater presence of non-financial corporate bonds in domestic bond markets. Columns (1)-(3) show that higher inequality is significantly and positively correlated with the share of outstanding bonds issued by non-financial corporations.¹¹

In Columns (1) and (2), a one-unit increase in the S80/S20 ratio of total population and working age population, capturing a more top-heavy income distribution, is associated with a 0.69 and 0.65 percentage point rise in the corporate share of domestic bonds, significant at the 5% and 10% level, respectively. Alternative measures of overall inequality yield consistent results. Column (3) shows that a 0.1-point increase in the Gini coefficient of disposable income (post taxes and transfers) for the total population is associated with a 2.56 percentage point rise in the corporate share of domestic bonds. Taken together, these results suggest that more unequal economies tend to exhibit shifts in their portfolios toward private, non-financial debt instruments. This finding is consistent with a supply-side mechanism in credit markets, as wealthier households capture a larger share of disposable income, they channel their savings into relatively safe yet higher-yielding corporate bonds, thereby reshaping the composition of domestic bond markets.

Column (4) shifts the focus from bond market composition to its overall size, using the total public bond market capitalization (public bonds as a share of GDP multiplied by current GDP in U.S. dollars) as the dependent variable.¹² The results indicate that higher income inequality, measured

¹¹Based on data availability, the sample covers annual observations from 2000 to 2021 for the following 18 countries: Australia, Brazil, Canada, China, Croatia, Denmark, Hungary, Iceland, Israel, Japan, South Korea, Mexico, Norway, Russia, South Africa, Sweden, Switzerland, and Turkey.

¹²Based on data availability, the sample covers annual data from 2000 to 2017 for the following 54 countries: Argentina, Australia, Austria, Belgium, Brazil, Canada, Chile, China, Colombia, Croatia, Cyprus, Czech Republic, Denmark, Finland, France, Germany, Greece, Hong Kong, Hungary, Iceland, India, Indonesia, Ireland, Israel, Italy,

by the Gini coefficient before taxes and transfers, is positively and significantly associated with larger public bond markets. A 0.1 point increase in this Gini coefficient is linked to a US \$ 49.6 trillion rise in total bond market capitalization, significant at the 10% level. This finding suggests that greater income concentration at the top contributes not only to changes in the composition of financial markets but also to their overall expansion. One possible channel is that top-income households exhibit a higher marginal propensity to save, thereby increasing aggregate demand for fixed-income assets such as public bonds.

Table 2: Relation between non-financial corporate bonds and measures of income inequality

	(1)	(2)	(3)	(4)
	Corp. bonds %	Corp. bonds %	Corp. bonds %	Bond Mkt Cap
S80/S20 Total pop.	0.685** (0.330)			
S80/S20 Working age pop.		0.650* (0.344)		
GINI Total pop.			25.63* (13.74)	
GINIB Working age pop.				496.1* (257.0)
Observations	173	173	173	246
Country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Columns (1)-(3) use the ratio of the total amount of domestic non-financial corporate bonds and notes outstanding to the total amount of domestic bonds and notes outstanding as the dependent variable. The measure of liabilities to GDP includes all debt liabilities, such as bonds, debentures, notes, and money market or other negotiable debt instruments. All these variables are sourced from the Global Financial Development Database. Column (4) is public bond market capitalization in USD trillion, taken from the Financial Development and Structure Dataset (revised September 2019) and World Bank Database, as the dependent variable. Measures of income inequality are obtained from the World Inequality Database (WID), and information for the control variables (e.g., lagged GDP per capita, lagged liabilities-to -GDP and lagged inflation) is obtained from the World Bank. All regressions include country and time fixed effects.

Although these estimates are not intended to be interpreted as causal effects, they suggest a consistent and robust association between income inequality and the development of corporate bond markets. To address the possibility that this relationship reflects broader macro-financial conditions rather than inequality itself, the regressions control for several additional covariates.

Japan, South Korea, Luxembourg, Malaysia, Malta, Mexico, Netherlands, New Zealand, Norway, Pakistan, Peru, Philippines, Poland, Portugal, Russia, Saudi Arabia, Singapore, Slovak Republic, Slovenia, South Africa, Spain, Sweden, Switzerland, Thailand, Turkey, United Kingdom and United States.

GDP per capita is included to capture the role of economic development, since richer economies typically exhibit more sophisticated financial systems and deeper capital markets. Liabilities-to-GDP proxies for the overall degree of financial depth and indebtedness, which could affect the relative space for corporate versus other forms of debt. Inflation is added as a measure of macroeconomic stability, as high or volatile inflation tends to erode investor demand for long-term fixed-income instruments and may hinder bond market development. Together, these controls help isolate the relationship between inequality and corporate bond markets from other structural or macroeconomic drivers.

This raises a key question: Is income inequality more closely associated with firms' debt structure, the composition of borrowing between banks and bond markets, rather than their capital structure, traditionally defined by the trade-off between debt and equity? While classical corporate finance emphasizes capital structure decisions, recent trends suggest a shift in the relevant margins of adjustment. In particular, U.S. initial public offerings (IPOs) activity has declined sharply, with many firms delaying or avoiding equity issuance. For instance, [Doidge et al. \(2013\)](#) documents that between the 1990s and 2000s, IPO activity in the U.S. fell by 8%, even as IPO capital raised outside the U.S. grew by 65%. Also, [Gao et al. \(2013\)](#) shows that from 1980 to 2000, an average of 310 firms went public each year in the U.S., whereas since 2000, the average has dropped to just 99 IPOs annually, with the decline particularly pronounced among smaller firms.

In contrast, there has been a marked increase in corporate bond issuance. The average issue size in the U.S. rose from approximately USD 479 million during the 2000-2007 period to USD 837 million between 2008 and 2018. Over the same time frame, the total volume of debt raised through corporate bond markets doubled, from roughly USD 300 billion in 2000 to USD 600 billion in 2018 [OECD \(2019\)](#). These trends suggest that the composition of debt, between bank loans and bonds, has become an increasingly important margin of firm financing, potentially influenced by shifts in household savings behavior and broader macro-financial dynamics.

Discussion about the causes of the permanent rise in income inequality: The existing literature offers several explanations for the persistent rise in income inequality in the U.S. over the past few decades. Key factors highlighted are: the skill-biased technological change indicates that the demand for highly educated workers has outpaced the supply, thereby widening the wage gap between college-educated and non-college-educated workers ([Goldin and Katz, 2008](#))¹³; the rise of self-employment, entrepreneurship, and the concentration of income among business owners has contributed to top-end inequality ([Smith et al., 2019](#)); the advances in information technology and

¹³However, [Heathcote et al. \(2010a\)](#) document that wage differentials by education, age, and race have stabilized since 2000, while the gender gap has continued to narrow.

automation have displaced routine jobs, compressing middle-skill wages while raising demand for both high-skill workers, who complement technology, and low-skill service workers (Autor et al., 2006); the rise of import competition, particularly from China since the early 2000s, has depressed wages in manufacturing and other tradable sectors, widening income dispersion across regions and occupations (Autor et al., 2013); and the declining unionization, lower real minimum wages, and changes in top marginal tax rates have also amplified wage and after-tax income inequality (Card, 1998; Piketty and Saez, 2003).

These mechanisms help explain the persistent rise in U.S. wage dispersion in recent decades. In Sections 3, 4 and 5, however, my analysis abstracts from these multiple channels and interprets the increase in income inequality as arising exclusively from higher dispersion in idiosyncratic labor productivity shocks.

2.2 Heterogeneous composition of liquid wealth across households

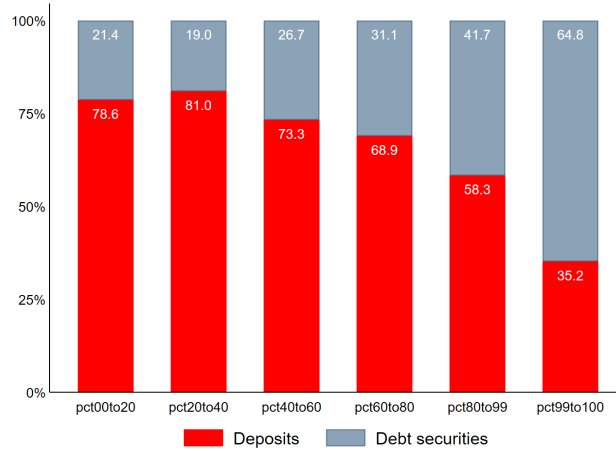
The period from the late 1970s to the mid-1990s witnessed significant technological advances in the issuance of debt securities, including the rise of securitization and the development of private debt markets (Buchak et al., 2024). These innovations contributed to a marked reduction in informationally sensitive lending. In parallel, financial innovation also reached retail markets, expanding household access to a broader range of financial products.¹⁴

Despite these advances, households continue to exhibit considerable inertia in rebalancing their portfolios, even between highly liquid assets such as deposits and bonds. In a frictionless environment without portfolio adjustment costs, all households would shift entirely toward the asset offering the highest yield. In reality, however, households across the income distribution hold a mix of assets, reflecting frictions, preferences, and constraints that limit such reallocation.

Figure 2 reveals heterogeneity in the composition of household liquid assets across income percentiles. Using data from the Distributional Financial Accounts (DFA), I compute liquid wealth for each percentile in each year as the sum of deposits and bond holdings, where bond holdings include three categories: corporate and foreign bonds, other loans and advances, and debt securities. I then calculate deposit shares as a percentage of total liquid wealth and average the composition across percentiles from 1989 to 2019. The results show that lower-income households (bottom 20%) allocate nearly 80% of their liquid assets to deposits, while the deposit share declines sharply with income. In contrast, households in the top 1% allocate only nearly one-third of their liquid wealth

¹⁴For instance, the rise of online brokerage platforms, bond ETFs, and reduced trading costs.

Figure 2: Composition of liquid wealth across income percentiles



Notes: Liquid wealth is defined as the sum of deposits and bond holdings (excluding U.S. and municipal securities). The red and blue bars display the average share of deposits and bonds as a percentage of total liquid wealth, respectively, across income percentiles from 1989 to 2019 based on data from the Distributional Finances Accounts (DFA).

to deposits, with the remaining two-thirds held in debt securities such as corporate bonds. This systematic shift suggests that higher-income earners exhibit a stronger preference for higher yields but less cash-like instruments, while lower-income earners remain disproportionately reliant on deposits. These differences in portfolio composition underscore the relevance of modeling household heterogeneity in liquidity demand when examining credit market dynamics.

On the other hand, while life-cycle motives affect total wealth accumulation, survey evidence shows relatively uniform liquid saving behavior across age cohorts. Figure A5 shows that the ratio of deposits to bond savings varies modestly across age groups, particularly among older cohorts (40 years old and above). While younger households consistently exhibit higher deposit reliance, the overall cross-age variation is small compared to the inequality observed across the income distribution. This suggests that age-cohort dynamics alone cannot explain the substantial heterogeneity in liquidity holdings, motivating a focus on inequality and portfolio sorting across the income dimension.

2.3 The Downward Trend of the Corporate Liquidity Premium

In this paper, the corporate liquidity premium is defined as the spread between the interest rate on corporate bonds and the return on deposits. Because consistent historical data on deposit

rates are unavailable for the full period of analysis, I instead proxy this spread using the difference between corporate bond yields and bank loan rates of comparable maturity and credit quality. To construct this measure, I use aggregate data on returns for corporate bonds and commercial paper from the FRED database, together with firm-level information on the costs of bond and bank loan transactions from Capital IQ, employing the categorization of transactions explained in the Subsection 2.1. The resulting two spreads, adjusted by inflation, are shown in Figure 3. The red line shows the three-year moving average of the return differential between high-rated corporate bonds with maturities of one to two years and 3-month commercial paper issued by AA-rated firms. Although commercial papers are not bank loans, [Allen et al. \(2023\)](#) describes that commercial papers offer large firms the ability to raise short-term debt as an alternative to bank loans.

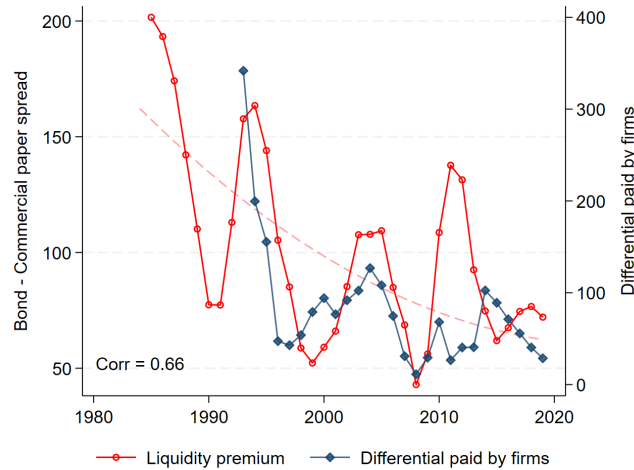
Complementing this, the blue line uses firm-level data to show the three-year moving average cost differential between bond and bank financing for comparable maturities. In this exercise, I group bond and bank loan transactions by maturity for each year from 1990 to 2019. I measure maturity as the number of years between the issuance and maturity dates, dropping transactions with undefined maturities or maturities exceeding 100 years. I then create maturity bins and I assign each credit transaction to one specific bin.¹⁵ Both the aggregate and firm-level series display a clear downward trend over time.

I interpret the red line as a proxy for the liquidity premium in corporate credit markets. Because firms issue the bonds in question with minimal credit risk, and despite having longer maturities than commercial papers, they are highly liquid and actively traded in secondary markets. Thus, the observed spread is more likely to reflect compensation for lower immediacy or liquidity rather than for default risk.

One concern is that the Treasury term premium in Figure 3 could confound the red line pattern, which incorporates expectations about future interest rates and compensation for macroeconomic risk. Appendix Figure A6 adjusts for this by subtracting the Treasury yield curve slope (e.g., the difference between one-year and three-month Treasury rates), isolating the component of the spread that more directly reflects credit and liquidity risk. The adjusted liquidity premium becomes noticeably flatter but still trends downward.

¹⁵I create bins of maturities such as bin 1 includes instruments with 1-2 years to maturity, bin 2 covers 3-5 years, bin 3 covers 6-10 years, and so on, with the final bin (bin 9) containing instruments with maturities greater than 60 years.

Figure 3: Spread of corporate bonds - commercial papers and differential cost paid by firms



Notes: The solid red line plots the three-year moving average of the difference between the annual average yield on corporate bonds rated AAA, AA, or A with maturities of one to two years, and the interest rate on 3-month commercial papers offered to AA-rated firms, both adjusted by inflation, based on FRED data. The red dashed line plots the quadratic trend of this spread. The blue line displays the three-year moving average of the difference between the average cost of funds from corporate bond issuance and bank loans, constructed using firm-level maturity data from Capital IQ.

While the bond-commercial paper spread does not directly measure household returns, it reflects equilibrium conditions in financial markets that influence the pricing of liquidity across the entire economy. In particular, a decline in this spread suggests that the compensation required to hold less liquid assets has fallen, which may lower the effective return differential between instruments available to households, such as deposits and bonds. This supports the idea that improvements in corporate bond market liquidity and intermediation efficiency can indirectly shape household portfolio choices by compressing liquidity premium and altering the relative attractiveness of financial assets.

Although disentangling supply and demand remains empirically challenging, [Becker and Ivashina \(2014\)](#) show that firms substitute away from bank loans and toward bond financing when credit conditions tighten or banks face stress. Similarly, [Zhu \(2021\)](#) find that bondholder inflows predict lower likelihoods of equity issuance and bank loan initiation, further underscoring the substitutability and interconnectedness of credit instruments.

The next section develops a quantitative model consistent with the empirical patterns documented and the core mechanism of the paper. Rising income inequality shifts the composition of household savings, concentrating bond demand among higher-income households and leaving lower-income

households more dependent on deposits (Subsection 2.2). Since financial intermediaries rely on household savings, this reallocation reshape the aggregate supply of intermediary capital, toward bonds. As a result, firms rely more on bond financing (Subsection 2.1), and the equilibrium liquidity premium declines (Subsection 2.3).

3 Quantitative model

In this section, I build a dynamic general equilibrium model with long-lived heterogeneous households operating in incomplete markets and heterogeneous firms that optimally choose their corporate debt structure. The economy features three types of agents: households, financial intermediaries, including a representative bank and bond markets, and productive firms.

3.1 Households

Time is discrete and infinite. There is a unit mass of households indexed by $i \in [0, 1]$. Households live infinitely, have preferences over consumption, and discount the future at rate β . They supply labor inelastically but experience shocks to their efficiency units of labor z_{it} . They have access to two financial assets: corporate bonds b_{it} and deposits d_{it} . The government collects labor income taxes at a rate τ^g , so the after-tax labor income is $(1 - \tau^g) W_t z_{it}$.

In reality, bonds offer higher return but are riskier than bank deposits, due to price fluctuations. To capture this trade-off in a tractable way, I assume that whenever a household adjusts its holdings of bonds, it needs to pay a stochastic adjustment cost χ_{it} that follows a logistic distribution with mean μ_χ and scale σ_χ . Because of this assumption, from hereon, I refer to bonds as illiquid and deposits as liquid. Following Bayer et al. (2019), this formulation ensures that the first-order conditions in a model with fixed adjustment probabilities are equivalent to those in a model with state-dependent adjustment probabilities. The presence of these adjustment costs induces households to optimally trade off the marginal benefit and cost of rebalancing their bond portfolios, resulting in infrequent and lumpy adjustment behavior.¹⁶

Finally, to capture the empirical fact that households at the bottom of the wealth distribution still hold a positive amount of bond assets, I introduce liquidity constraints. I assume that holdings of liquid assets have a lower bound $-\underline{d}$, and holdings of illiquid assets have to be non-negative.

¹⁶These costs can capture a wide range of real-world frictions, such as risk premia, transaction fees, information processing constraints, inattention, or regulatory barriers, that discourage frequent portfolio reallocations.

Specifically, households are allowed to access unsecured credit to finance consumption and/or increase bond holdings when portfolio adjustment is permitted. This credit incurs an additional cost $\hat{r} > 0$, which generates a mass of households with zero unsecured borrowing but the potential to take on debt.¹⁷

Then, taking prices as given, households will maximize:

$$\begin{aligned} & \max_{\{c_{it}, b_{it+1}, d_{it}\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}) - \mathbb{I}_{\Delta b_{it} \neq 0} \chi_{it} \right] \\ \text{s.t. } & c_{it} + d_{it+1} + b_{it+1} = (1 - \tau^g) W_t z_{it} + (1 + r_t^d) d_{it} + (1 + r_t^b) b_{it} + \Pi_t \\ & d_{it} \geq -\underline{d}, \quad b_{it} \geq 0 \\ & z_{it} = \frac{\tilde{z}_{it}}{\int \tilde{z}_{it}} \text{ where } \tilde{z}_{it} = \exp(\rho \log \tilde{z}_{it-1} + \varepsilon_{it}), \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma_t) \end{aligned} \quad (3)$$

where instantaneous utility is $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ with $\gamma \geq 0$. Labor income follows a log-AR(1) process, and each household receives a fixed share of firms' dividends, Π_t .¹⁹ Also, the interest rate of deposit is defined by:

$$r_t^d = \begin{cases} r_t^d + \hat{r} & , d_{it} < 0 \\ r_t^d & , d_{it} \geq 0 \end{cases} \quad (4)$$

Furthermore, the policy functions of the household's problem will be nonlinear functions of the state variables (d_t, b_t, z_t) and joint distribution $\Theta_t := G(d_t, b_t, z_t)$.

Next, I present this problem in recursive form, as a function of the idiosyncratic state variables (d, b, z) and the aggregate distribution of households Θ . The dynamic problem of households is characterized by two Bellman equation: V_a when bond holdings can be adjusted and V_n when they cannot.

When adjustment is permitted, households optimally allocate between deposits d'_a and bonds b'_a , with the resulting asset positions determining the continuation value, defined as the expected value of future welfare.

¹⁷This cost could represent an intermediation fee for banks, reflecting the need to monitor unsecured borrowers to prevent default.

¹⁹This share is proportional to the household's labor productivity and cannot be traded as an asset.

$$\begin{aligned}
V_a(d, b, z|\Theta) &= \max_{d'_a, b'_a} u(c(d, d'_a, b, b', z)) + \beta \mathbb{E}V(d', b', z'|\Theta') \\
\text{s.t. } c + d'_a + b'_a &= (1 - \tau^g)Wz + (1 + r^d)d + (1 + r^b)b + \Pi(z) \\
d'_a &\geq -\underline{d}, \quad b'_a \geq 0 \\
\mathbb{E}V(d', b', z|\Theta) &= \mathbb{E}_{\chi', z'} [\max \{V_a(d', b', z'|\Theta) - \chi', V_n(d', b', z'|\Theta)\}]
\end{aligned} \tag{5}$$

$$\begin{aligned}
V_n(d, b, z|\Theta) &= \max_{d'_n} u(c(d, d'_n, b, b, z)) + \beta \mathbb{E}V(d', b, z'|\Theta') \\
\text{s.t. } c + d'_n &= (1 - \tau^g)Wz + (1 + r^d)d + r^b b + \Pi(z) \\
d'_n &\geq -\underline{d} \\
\mathbb{E}V(d', b', z|\Theta) &= \mathbb{E}_{\chi', z'} [\max \{V_a(d', b', z'|\Theta) - \chi', V_n(d', b', z'|\Theta)\}]
\end{aligned} \tag{6}$$

In contrast, when adjustment is not permitted, households optimally choose only their deposit holdings d'_n , so the continuation value depends exclusively on variations in d'_n . Moreover, a household will choose to pay the fixed cost to adjust its portfolio if and only if the value function conditional on adjusting (V_a), net of the adjustment cost (χ'), is greater than or equal to the value function conditional on not adjusting (V_n): $V_a(d', b', z'|\Theta) - \chi' \geq V_n(d', b', z'|\Theta)$. The adjustment cost χ is drawn from a cumulative distribution function F_χ with mean μ_χ and scale σ_χ . The probability of adjustment is therefore given by:

$$\begin{aligned}
\chi' &\leq V_a(d', b', z'|\Theta) - V_n(d', b', z'|\Theta) \\
\nu^*(d', b', z'|\Theta) &:= F_\chi[V_a(d', b', z'|\Theta) - V_n(d', b', z'|\Theta)]
\end{aligned} \tag{7}$$

Conditional on paying the adjustment cost, a household chooses a portfolio that balances the higher liquidity of deposits, useful for smoothing consumption, against the higher return offered by bond savings in equilibrium, which is a feature of calibration.

Equation 7 implies that even for states where $\Delta V \equiv V_a - V_n$ is small, there is still a nonzero probability of adjustment, and for states with large ΔV , adjustment is highly likely but not certain. This stochastic formulation replaces the sharp "inaction regions" typical of models with deterministic fixed adjustment costs. Here, the cutoff cost $\chi^*(d, b, z) = \Delta V(d, b, z)$ defines the threshold at which a household is indifferent between adjusting and not. The CDF of χ gives the adjustment probability, so it rises smoothly as ΔV increases. Because the model integrates over all possible realizations of χ when solving for policies, the policy and value functions are smooth in (d, b, z) .

Euler Equations: The policies of consumption, deposit, and bond holdings are denoted by c_i^* , d_i^* , and b_i^* , respectively, where $i \in \{a, n\}$. The first-order conditions for an interior solution of the households' problem are:

$$[b^*] \frac{\partial u(c_a^*)}{\partial c} = \beta \mathbb{E} \left[\nu \frac{\partial u(c_a^*(d_a^*, b^*, z'))}{\partial c} (1 + r^b) + (1 - \nu) \frac{\partial V_n(d_a^*, b^*, z')}{\partial b} \right] \quad (8)$$

$$[d_a^*] \frac{\partial u(c_a^*)}{\partial c} = \beta \mathbb{E} \left((1 + r^d) \left[\nu \frac{\partial u(c_a^*(d_a^*, b^*, z'))}{\partial c} + (1 - \nu) \frac{\partial u(c_n^*(d_a^*, b^*, z'))}{\partial c} \right] \right) \quad (9)$$

$$[d_n^*] \frac{\partial u(c_n^*)}{\partial c} = \beta \mathbb{E} \left((1 + r^d) \left[\nu \frac{\partial u(c_a^*(d_n^*, b, z'))}{\partial c} + (1 - \nu) \frac{\partial u(c_n^*(d_n^*, b, z'))}{\partial c} \right] \right) \quad (10)$$

Derivations are provided in Appendix B.1. These conditions imply that when a household decides between holding deposits and bond savings, it must compare three margins. First, the one-period return differential $r^b - r^d$, weighted by the marginal utility of consumption when adjustment is allowed. Second, the return on deposits when adjustment is not possible, given by $(1 + r^d) \frac{\partial u(c_n^*(d_a^*, b^*, z'))}{\partial c}$. Third, the marginal value of bond holdings in the no-adjustment state, $\frac{\partial V_n(d_a^*, b^*, z')}{\partial b}$, which captures both the utility from the bond's return stream and the option value of selling bonds in future periods when adjustment becomes available.

3.2 Firms

Departing from the standard RBC literature, I introduce idiosyncratic risk to the firms' stock of assets that directly affects production. This approach, as in Villacorta (2018), allows me to decompose firms' collateral into two components: (i) a riskless portion, whose value is preserved regardless of whether an adverse shock hits the firm, and (ii) a risky portion, whose value is directly affected when the firm experiences such a shock.

There is a unit mass of productive firms indexed by $j \in (1, 2]$. Firms share a common technology A . They are heterogeneous along two dimensions: initial net worth n_{jt} and the realization of an idiosyncratic productivity shock $\omega_{jt} \in \{0, 1\}$ that follows a Bernoulli process with probability p_H that represents the fraction of firms that succeed in their projects. Whenever there is an adverse shock $\omega_{jt} = 0$, there is a partial loss of capital $\kappa > 0$, and the firm can't produce in period t . This shock is privately observed. Otherwise, each firm produces a homogeneous good y_{jt} according to the production function:

$$f(A_t, k_{jt+1}, l_{jt}, \omega_{jt}) = \omega_{jt} A_t k_{jt+1}^\alpha l_{jt}^{1-\alpha} \quad (11)$$

where the factors of production are capital k_{jt+1} and labor l_{jt} .²⁰ Capital depreciates every period at rate δ and firms hire workers before the realization of ω_{jt} , but wage payments occur after. I assume that wages are contingent on firm productivity to prevent firms from operating at a loss when $\omega_{jt} = 0$. Also, there is no reallocation of workers after the realization of ω_{jt} , preventing any labor adjustments that could distort the return on capital. The labor market is competitive, with firms taking w_t as given, and wages falling to zero if the firm fails.

3.2.1 Investment block

After determining the optimal wage contracts, individual firms take wages as given, so their gross profits depend on wages and are linear in their own capital.²¹ However, because wages are determined by aggregate capital, a firm's marginal profit with respect to capital ultimately depends on the aggregate capital stock, implying decreasing returns. The marginal return on capital for each firm is therefore given by:

$$R_{jt}^K = \begin{cases} \alpha A K_{t+1}^{\alpha-1} + 1 - \delta & \text{with } p_H \\ (1 - \delta)(1 - \kappa) & \text{with } 1 - p_H \end{cases} \quad (12)$$

From an aggregate perspective, the return on capital decreases with K_{t+1} , but from an individual firm's perspective it is constant. Thus, if all firms simultaneously expand their assets, the aggregate return on capital falls, and competition for a fixed supply of labor drives wages up. Finally, since there are decreasing returns to aggregate capital, its level must be sufficiently high to ensure that firms pay the return expected by households, rather than the reverse. Therefore, a crucial assumption is that the expected return on capital exceeds one.

Assumption 1: $\mathbb{E}_t R_{jt}^K \equiv R_t^K > 1$. Return on capital is high enough to incentivize households to save.

²⁰In this section, the terms "capital" and "assets" are used interchangeably.

²¹Details of the labor market equilibrium are provided in Appendix C.1.

3.2.2 Financing block

In the absence of financial frictions, firms would finance themselves using the cheapest available instrument, and any cost differential between bonds and bank loans would be eliminated by competition in the financial sector. Consequently, a positive liquidity premium, defined as the wedge between less liquid bonds and more liquid deposits, would not persist in equilibrium, as arbitrage by firms and financial intermediaries would equalize marginal costs of funding. To capture realistic debt market dynamics, my model introduces financial frictions due to moral hazard in the spirit of [Holmstrom and Tirole \(1997\)](#).

The idiosyncratic productivity shock ω_{jt} gives rise to two sources of financial frictions. On the one hand, it generates repayment risk, since production ceases when the shock hits a firm. On the other hand, information asymmetries create incentives for firms to abscond profits. Also, firms have access to both bank loans and bond financing, and choose their debt composition to maximize expected net worth, subject to contractual arrangements that mitigate these frictions.

Firms raising external funds can pledge their entire stock of assets as collateral, allowing both banks and bondholders to hold claims on the firm's assets. However, the enforceability of financial contracts determines the priority structure of these claims, ensuring that bondholders are repaid first while banks bear the riskier portion of debt.

Also, firms report the realization of ω_{jt} independently to banks and bondholders. Since bondholders lack monitoring capabilities, they cannot verify the firm's true state, creating a moral hazard problem that restricts bond financing to risk-free debt. Hence, the amount of risk-free bonds that firms can issue is limited by the safe part of their capital, $1 - \kappa$. Then, the riskless borrowing constraint of bond issuance is defined by:

$$R_t^B L_{jt}^B \leq (1 - \delta)(1 - \kappa)k_{jt+1} \quad (13)$$

In contrast, banks can monitor firms, allowing them to accept risky debt while enforcing incentive constraints to deter misreporting. This separation of reporting justifies the different financial contract structures for bondholders and banks. As a result, firms have incentives to misreport the true realization of ω_{jt} and divert resources for private benefits.

Furthermore, banks accept risky debt limited by their monitoring effectiveness $\phi \in (0, 1)$. Unlike bondholders, banks can monitor firms, allowing them to extend risky debt that is not fully collateralized. To prevent misreporting, banks impose an incentive compatibility constraint, ensuring

that firms truthfully report their outcomes rather than diverting funds. Appendix D.1 describes the constraints on financial contracts assumed in the model.

In the absence of enforcement, firms can divert profits for private gain. Banks expand external financing while imposing discipline that curbs such behavior. Banks mitigate misreporting through monitoring, which acts as an enforcement device, restricting firms from diverting a fraction ϕ of the risky value of capital. Since banks assume the riskier part of the firm's liabilities, they only accept risky debt up to a fraction ϕ of the risky capital that can be effectively monitored. The stochastic loan rate promised to banks at t is denoted by:

$$R_{jt}^D = \begin{cases} R_t^D & \omega_{jt} = 1 \\ 0 & \omega_{jt} = 0 \end{cases} \quad (14)$$

where ω_{jt} denotes the realization of the idiosyncratic productivity shock. The zero recovery value for the bank loan is a normalization: given the binomial nature of the idiosyncratic shock, any loan with a positive recovery value can be decomposed into a risk-free component and a risky recovery debt instrument, as long as they satisfy the feasibility constraints.

Given the loan rate R_t^D , the total promise from the risky bank loans issued by firms is limited by:

$$R_t^D L_{jt}^D \leq \phi \kappa (1 - \delta) k_{jt+1} \quad (15)$$

Firms begin period t with initial net worth n_{jt} and can raise additional financing from both households and banks by issuing either risk-free bonds or risky bank loans. I denote with L_{jt}^B the amount of funds borrowed through risk-free bonds and L_{jt}^D the funds obtained via risky bank loans. Total capital invested by firm j at period t is:

$$k_{jt+1} = n_{jt} + L_{jt}^B + L_{jt}^D \quad (16)$$

and combining Equations 13, 15 and 16, I get

$$k_{jt+1} \leq \frac{1}{\left(1 - \frac{(1-\delta)(1-\kappa)}{R_t^B} - \frac{\phi \kappa (1-\delta)}{R_t^D}\right)} n_{jt} \quad (17)$$

where R_t^B and R_t^D are returns that clear the market of risk-free bonds and risky bank loans, respectively.

Firm's leverage: It is important to notice that Equation 17 imposes a maximum leverage of firms that depends on their initial net worth and the relative repayment (per unit) to financial intermediaries. Moreover, lower interest rates and greater monitoring effectiveness (ϕ) enhance firms' ability to expand their capital stock by increasing their borrowing capacity.

3.2.3 Firm's problem

Firms will maximize their net worth and only will pay dividends to households when they exit the market which occurs each period with probability τ . Firms have a discount factor equivalent to the stochastic discount factor of households denoted by Λ . Conditional on existing, firms start period t with an initial net worth n_{jt} , and they borrow from households and banks implying that the law of motion of net worth and capital after the realization of the idiosyncratic productivity shock are defined by:

$$n_{jt+1} = R_{jt}^K k_{jt+1} - R_t^B L_{jt}^B - R_{jt}^D L_{jt}^D \quad (18)$$

Thus, taken R_t^K , R_t^B and R_t^D as given, firms will choose optimally L_{jt}^B and L_{jt}^D to maximize their respective net worth as follows:

$$V_t^F(n_{jt}) = \max_{L_{jt}^B, L_{jt}^D} \tau \Lambda \mathbb{E}_t[n_{jt+1}] + (1 - \tau) \Lambda \mathbb{E}_t[V_{t+1}^F(n_{jt+1})] \quad (19)$$

subject to 13, 15, 16, and 18.

3.3 Financial intermediaries

There are two types of financial intermediaries: (i) bond markets composed of uninformed investors who invest in secured firm loans, and (ii) traditional banks, which are deposit-taking institutions that invest all household deposits in firm loans that are not necessarily secured. When idiosyncratic shocks hit firms and some become unable to repay, banks have the ability to pool risk across firms and ensure repayment to households. This risk-sharing capacity allows banks to offer deposit contracts that are safe from the perspective of households, despite the underlying exposure to firm default risk.

3.3.1 Bond markets

Bond markets consist of uninformed investors, treated here as a single entity with no specialized technology for monitoring or capital production. These investors pool household savings, offering a return of $1 + r_t^b$. Then, profits in the bond market are defined by $\pi_t^B = R_t^B L_t^B - (1 + r_t^b) B_t = R_t^B \int_j L_{jt}^B d_j - (1 + r_t^b) B_t$. Since the bond market is perfectly competitive, equilibrium implies $\pi_t^B = 0$. Moreover, as total bond issuance equals the resources collected from households, $L_t^B = B_t = \int_i b_{it} d_i$ leading to:

$$R_t^B = 1 + r_t^b \quad (20)$$

3.3.2 Banks

There is a continuum of competitive banks, treated as a single party that collects deposits from households at a cost of funding $1 + r_t^d$. Banks possess a specialized monitoring technology, which incurs a cost $\hat{c}$ for each unit of firm credit. This monitoring mitigates the moral hazard problem by improving borrower discipline. Banks are short-lived: they operate for one period, after which they are replaced by new banks with identical technology.

In the absence of unsecured consumer credit, bank profits are given by: $\pi_t^D = \tilde{R}_t^D L_t^D - (1 + r_t^d) D_t - \hat{c} L_t^D \geq 0$ and the average return on banks' portfolio loans is given by $\tilde{R}_t^D = \frac{\int_j R_{jt}^D L_{jt}^D d_j}{L_t^D}$. Since risk diversification is costless and by LLN then the average return per unit of capital is:

$$\tilde{R}_t^D = p_H R_t^D \frac{\int_j L_{jt}^D}{L_t^D} = p_H R_t^D$$

then $p_H R_t^D L_t^D \geq (1 + r_t^d) D_t + \hat{c} L_t^D$ and since banks have no net worth $L_t^D = D_t$ then the aggregate supply of bank loans is defined by:

$$p_H R_t^D = 1 + r_t^d + \hat{c} \quad (21)$$

Equation 21 remains unchanged when unsecured consumer credit is introduced.²²

²²Details on the introduction of unsecured consumer credit are provided in Appendix D.

3.4 Aggregation

The aggregate output and the stock of assets are defined by:

$$Y_t = p_H A K_t^\alpha \quad (22)$$

$$K_{t+1} = N_t + L_t^B + L_t^D \quad (23)$$

As firms exit over time with probability τ , I assume that every period a new set of firms enters and replaces them so that the set of firms remains $\mathbb{J} = (1; 2]$. Every period, a measure τ of firms enters the market. Each new firm is endowed with a random initial capital with a mean equal to the fraction ζ of the average net worth of existing firms. By aggregating Equation 18 and considering the entry and exit of firms, the initial aggregate net worth in $t + 1$ is defined by:

$$N_{t+1} = (1 - \tau + \tau\zeta) \left(R_t^K K_{t+1} - R_t^B L_t^B - p_H R_t^D L_t^D \right) \quad (24)$$

where the return on capital (per unit) R_t^K and the average bank loan repayment rate (per unit):

$$R_t^K = p_H \alpha A K_{t+1}^{\alpha-1} + p_H (1 - \delta) \kappa + (1 - \delta)(1 - \kappa) \quad (25)$$

3.5 General equilibrium

A recursive equilibrium is a set of households' policy functions $\{c_i^*, d_i^*, b_i^*, \nu_i^*\}, \forall i \in \mathbb{I}$, firms' policy functions $\{l_j^*, k_j^*, L_j^{*B}, L_j^{*D}, n_j^*\}, \forall j \in \mathbb{J}$, set of value functions of households $\{V_i^a, V_i^n\}, \forall i \in \mathbb{I}$, firms $\{V_j^F\}, \forall j \in \mathbb{J}$, banks and investors policy functions $\{R^D, R^B\}$, prices $\{r^d, r^b, w, R^K\}$, and distributions Θ over individual asset holdings such that:

- Given prices $\{r^d, r^b, w\}$, the value functions $\{V_i^a, V_i^n\}$, distributions Θ and policy functions $\{c_i^*, d_i^*, b_i^*, \nu_i^*\}$ solve the households' problem.
- Given $\{R^D, R^B, w, R^K\}$ and initial net worth n_j , the policy functions $\{l_j^*, k_j^*, L_j^{*B}, L_j^{*D}\}$ solve the firms' problem.
- Given $\{r^d, r^b, D_t, B_t\}$, policy function $\{\tilde{R}^D, R^B\}$ solves the banks and bond markets problem.

- The market of goods and labor clears:

$$\int_i c_i^* d_i = \int_j y_j^* d_j \quad (26)$$

$$\int_j l_j^* d_j = 1 \quad (27)$$

- The risk-free bond market and risky bank loans clears:

$$\int_i b_i^* d_i = \int_j L_j^{*B} d_j \quad (28)$$

$$\int_i d_i^* d_i = \int_j L_j^{*D} d_j \quad (29)$$

- Aggregate output, the law of motion of capital and net worth are defined by Equations 22, 23 and 24, respectively.

3.6 Equilibrium characterization

Using the numerical methods outlined in Appendix F, this section presents the policy functions of households and firms separately, and then describes their interaction in credit markets, which ultimately shapes the quantitative results discussed in Section 5.

In terms of portfolio allocation, households choose only their optimal deposit holdings when bond adjustment is not permitted, and jointly choose deposits and bonds when adjustment is allowed. Figure A7 compares the difference in optimal deposit shares between the most and least productive households under both cases. The contrast between the two panels shows that allowing portfolio adjustment amplifies heterogeneity in portfolio responses. When households cannot adjust, deposit shares vary little across productivity levels, reflecting rigid portfolios and muted precautionary behavior. Once adjustment is allowed, however, high-productivity but low-wealth households increase their liquidity buffers, while wealthier shift toward higher-yield bonds. As a result, the dispersion in deposit shares widens substantially, highlighting how portfolio flexibility amplifies the heterogeneity of saving responses to income risk.

Moreover, Figure A8 illustrates how productivity heterogeneity translates into differences in consumption and deposit share allocation. The differences in consumption (between the high and

low-productive households) have the strongest effects at the lowest wealth level. This gap narrows as deposits and bond holdings increase, reflecting the diminishing marginal utility of consumption and the ability of wealthier households to self-insure against income risk.

On the other hand, the largest differences in optimal deposit shares appear at low levels of wealth, suggesting that more productive but liquidity-constrained households hold larger deposit buffers to smooth consumption against income risk. As wealth increases, these differences decline, since productive households gradually shift toward bonds to benefit from higher returns. This pattern reflects the interaction between income risk, wealth accumulation, and portfolio composition. Productivity differences matter most when wealth is low, then precautionary motives are strongest and the incentive to save in liquid form is higher. However, as households accumulate assets, portfolio composition rebalances toward bonds, particularly for those with initially high deposit holdings or moderate bond positions.

Turning to firms' financing decisions, after deriving the credit demand functions for risk-free bonds and risky bank loans, the resulting policy functions are linear in firms' initial net worth. Consequently, all firms behave as a representative firm that takes the return on capital R_t^K , the risk-free bonds rate R_t^B and risky bank loan rate R_t^D as given.²³ When the interest rate on one of the credit sources is sufficiently low, the representative firm maximizes borrowing from that source until the corresponding collateral constraint binds. To further expand external financing and increase the stock of productive assets available for the next period, the firm then borrows through the alternative credit channel.

When income inequality rises, households strengthen their precautionary motives and increase savings to insure against higher income risk. This raises the overall supply of loanable funds but also changes its composition. Lower- and middle-income households expand deposits, while high-income households shift more into bonds. As a result, the aggregate supply of bonds grows faster than that of deposits. The excess demand for risk-free bonds pushes their yield down, prompting firms to substitute toward this cheaper source of funding until collateral constraints bind. Remaining financing needs are met through bank loans at higher rates. Consequently, for a given monitoring cost $\hat{c}$, the corporate liquidity premium compresses. The following subsections detail these mechanisms by examining the separate responses of households and firms and their general equilibrium interaction.

²³Details of the firm's dynamic problem are provided in Appendix C.2.

3.6.1 Aggregate savings and income inequality

This part examines how aggregate savings and their composition evolve as income inequality increases, holding equilibrium prices constant. Figure A9 shows how rising idiosyncratic labor productivity risk affects both the level and composition of household savings in partial equilibrium. As the variance σ increases and prices (i.e., r^d, r^b, W) are fixed, aggregate savings rise steadily, consistent with households engaging in precautionary saving to buffer higher income uncertainty.

By contrast, the deposit share of liquid wealth (red dots, right axis) also rises overall, indicating a portfolio reallocation toward the more liquid assets, but the pattern is not strictly monotonic. Instead, it exhibits small swings across values of σ . This non-monotonicity reflects the uneven portfolio responses of households. As income risk rises, the value of flexibility increases, even at the cost of accepting lower returns on deposits relative to illiquid assets (the liquidity premium is $r^b - r^d = 0.0565$). Thus, while higher income risk generally amplifies saving and strengthens demand for deposits, it does so in a lumpy fashion that reshapes the liquidity structure of household portfolios and, in turn, affects credit market dynamics.

3.6.2 Credit regimes

The firm's state space over bank loans and bonds (L_t^D, L_t^B) can be divided into two regions: (i) an unconstrained credit regime, where neither borrowing constraint binds, and (ii) a constrained credit regime, where one or both borrowing constraints are binding. Owing to the linearity of the firm's borrowing policies, the distribution of firms' net worth is irrelevant; hence, the analysis in the following sections abstracts from this distribution.²⁴

Unconstrained credit regime: If the returns across all funding sources are equalized such that $R_t^K = R_t^B = p_H R_t^D$, then the borrowing constraints are not binding. Also, in the steady state, firms' net worth satisfies $(1 - \tau)R^K = 1$, which ensures that firms do not accumulate net worth indefinitely and remain near the steady state. Under this regime, firms are indifferent between funding sources, since each yields the same return on capital.

Corollary 1. Under Assumption 3.2.1, the condition $(1 - \tau)R_t^K = 1$, and given that the return on capital in 25 holds, the equilibrium stock of capital lies within the interval $K_t \in [\underline{K}, \bar{K})$ where the lower and upper bounds are defined as follows:

²⁴Details of the firm's dynamic problem and the derivation of the conditions that characterize these regimes are provided in Appendix C.2.

$$\underline{K} \equiv \left(\frac{\alpha A p_H}{\frac{1}{1-\tau} - p_H(1-\delta)\kappa + (1-\delta)(1-\kappa)} \right)^{\frac{1}{1-\alpha}}; \quad \bar{K} \equiv \left(\frac{\alpha A p_H}{1 - p_H(1-\delta)\kappa + (1-\delta)(1-\kappa)} \right)^{\frac{1}{1-\alpha}} \quad (30)$$

Constrained credit regime: When either or both borrowing constraints bind, the firm's debt composition becomes endogenous. Specifically, when the constraint on risky bank loans or on bonds is binding, the firm's first-order conditions yield the following expressions, respectively.

Lemma 1. Kink point on risk-free bond return under a tight risky bank loan constraint. Suppose that the bank constraint binds. Then, the kink point on risk-free bond return, $\bar{R}_t^B$, is defined by:

$$\bar{R}_t^B \equiv \frac{p_H(\alpha A K_{t+1}^{\alpha-1} + (1-\delta)\kappa) + (1-\delta)(1-\kappa - \phi\kappa)}{1 - \frac{\phi\kappa(1-\delta)}{R_t^D}} \equiv \frac{R_t^K - \phi\kappa(1-\delta)}{1 - \frac{\phi\kappa(1-\delta)}{R_t^D}} \quad (31)$$

Furthermore, if $R_t^B < \bar{R}_t^B$ implies that the marginal value of borrowing through risk-free bonds is positive, then the bond constraint binds.

Lemma 2. Kink point on risky bank loan return under a tight risk-free bond constraint. Suppose that the bond constraint binds. Then, the kink point on risky bank loan return, $\bar{R}_t^D$, is defined by:

$$p_H \bar{R}_t^D \equiv \frac{p_H(\alpha A K_{t+1}^{\alpha-1} + (1-\delta)\kappa)}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} \equiv \frac{p_H(R_t^K - (1-\delta)(1-\kappa))}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} \quad (32)$$

Furthermore, if $R_t^D < \bar{R}_t^D$ implies that the marginal value of borrowing through risky bank loans is positive, then the bank loan constraint binds.

3.6.3 Optimal borrowing decisions

The following propositions can thus characterize the firm's demand for risk-free bonds and risky bank loans.

Proposition 3. The firm's demand for risk-free bonds is defined by:

$$L_{jt}^B = \begin{cases} \bar{L}_{jt}^B & \text{if } R_t^B < \bar{R}_t^B \\ [0, \bar{L}_{jt}^B] & \text{if } R_t^B = \bar{R}_t^B \\ 0 & \text{if } R_t^B > \bar{R}_t^B \end{cases} \quad (33)$$

where $\bar{L}_{jt}^B = \frac{(1-\kappa)(1-\delta)}{R_t^B} \frac{1}{\left(1 - \frac{(1-\kappa)(1-\delta)}{R_t^B} - \frac{\phi\kappa(1-\delta)}{p_H R_t^D}\right)} n_{jt}$ and $\bar{R}_t^B = \frac{R_t^K - \phi\kappa(1-\delta)}{1 - \frac{\phi\kappa(1-\delta)}{R_t^D}}$.

Proposition 4. The firm's demand for risky bank loans is defined by:

$$L_{jt}^D = \begin{cases} \bar{L}_{jt}^D & \text{if } R_t^D < \bar{R}_t^D \\ [0, \bar{L}_{jt}^D] & \text{if } R_t^D = \bar{R}_t^D \\ 0 & \text{if } R_t^D > \bar{R}_t^D \end{cases} \quad (34)$$

where $\bar{L}_{jt}^D = \frac{\phi\kappa(1-\delta)}{R_t^D} \frac{1}{\left(1 - \frac{(1-\kappa)(1-\delta)}{R_t^B} - \frac{\phi\kappa(1-\delta)}{p_H R_t^D}\right)} n_{jt}$ and $\bar{R}_t^D = \frac{R_t^K - (1-\delta)(1-\kappa)}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}}$.

A firm's demand for external funds depends on the relative cost of bonds and bank loans, each characterized by a kink point that equates the effective return on capital to the respective borrowing cost. Proposition 3 reveals that when the bond rate R_t^B falls below its threshold $\bar{R}_t^B$, firms prefer bonds as the cheaper source of credit, up to the collateral limit $\bar{L}_{jt}^B$. Conversely, if R_t^B exceeds $\bar{R}_t^B$, bond demand drops to zero, while at equality, it becomes perfectly elastic. Proposition 4 reveals a symmetric logic applies to bank loans. When the loan rate R_t^D is below its threshold $\bar{R}_t^D$, firms borrow from banks up to the collateral cap $\bar{L}_{jt}^D$, when it exceeds the threshold, loan demand ceases; and when equal, it is perfectly elastic.

3.6.4 Market for risk-free bonds and risky bank loans

The aggregate demand for risk-free bonds is defined by:

$$L_t^B = \begin{cases} \frac{(1-\kappa)(1-\delta)}{R_t^B} \frac{1}{\left(1 - \frac{(1-\kappa)(1-\delta)}{R_t^B} - \frac{\phi\kappa(1-\delta)}{p_H R_t^D}\right)} N_t & \text{if } R_t^B < \bar{R}_t^B \\ [0, \frac{(1-\kappa)(1-\delta)}{R_t^B} \frac{1}{\left(\frac{1}{1-\delta} - \frac{1-\kappa}{R_t^B} - \frac{\phi\kappa}{p_H R_t^D}\right)} N_t] & \text{if } R_t^B = \bar{R}_t^B \\ 0 & \text{if } R_t^B > \bar{R}_t^B \end{cases} \quad (35)$$

whereas the supply of risk-free bonds is defined by:

$$R_t^B = 1 + r_t^b \quad (36)$$

Furthermore, the aggregate demand for risky bank loans is defined by:

$$L_t^D = \begin{cases} \frac{\phi\kappa(1-\delta)}{R_t^D} \frac{1}{\left(\frac{1}{1-\delta} - \frac{1-\kappa}{R_t^B} - \frac{\phi\kappa}{p_H R_t^D}\right)} N_t & \text{if } R_t^D < \bar{R}_t^D \\ [0, \frac{\phi\kappa(1-\delta)}{R_t^D} \frac{1}{\left(\frac{1}{1-\delta} - \frac{1-\kappa}{R_t^B} - \frac{\phi\kappa}{p_H R_t^D}\right)} N_t] & \text{if } R_t^D = \bar{R}_t^D \\ 0 & \text{if } R_t^D > \bar{R}_t^D \end{cases} \quad (37)$$

whereas the supply of risky bank loans is defined by:

$$p_H R_t^D = 1 + r_t^d + \hat{c} \quad (38)$$

Therefore, the aggregate demand for risky bank loans and risk-free bonds is jointly determined, as changes in the interest rate of one instrument affect the firm's leverage decisions and, consequently, the demand for both types of financing. Figure A10 illustrates the relationship between aggregate deposits, aggregate bonds, and firms' net worth and return on capital.

Figure A10A reveals that aggregate net worth is a non-monotonic function of the financial asset composition. At low levels of deposits and bonds, firms are largely unconstrained and indifferent between bond and loan financing, enabling them to rely more on their internal funding (net worth). However, as the economy accumulates liquid wealth beyond a certain threshold, mainly when concentrated in deposits, financial frictions begin to bind. In equilibrium, firms substitute away from bank loans toward bond financing as the relative cost of borrowing through banks rises.

The shift in financing composition, driven by the tightening of borrowing constraints, leads firms to accumulate more internal funds, thereby shaping the equilibrium path of the return on capital. Figure A10B exhibits the reaction of the return on capital to these dynamics. In unconstrained regions, the marginal return on capital increases due to restricted firm financing, whereas in the constrained region, capital markets clear at a lower return, reflecting the lower cost of external funding and higher accumulation of assets K_{t+1} . Notably, the point labeled "SS2" (the new stationary equilibrium with higher productivity risk) is located in a region with lower capital returns and higher reliance on external funding than "SS1" (the baseline scenario), highlighting the nontrivial impact of income risk on firm-side outcomes.

Thus, Figure A11 maps the economy's feasible allocations into constrained and unconstrained regimes. The blue dots denote combinations of aggregate deposits and bonds for which financial constraints (either on bonds or deposits) bind. The region labeled "Constrained Regime" includes allocations with relatively high bond supply, reflecting firms' increased reliance on bond financing

as the bond interest rate declines. However, despite lower borrowing costs, the economy enters a constrained regime because firms hit an upper limit on how much they can borrow through risk-free bonds, due to the collateral constraint. This forces firms to adjust their debt structure.

In contrast, the "Unconstrained Regime" appears when bonds and deposits are scarce (to the left of the blue region) or when there is not enough bond supply to lower borrowing costs (to the right of blue region). Finally, the combinations of deposits and bonds inside the gray region reflect the unfeasible area where the value of collateral of risk-free bonds or risky bank loans are not high enough to cover the resources collected by bond markets and banks. Importantly, the equilibrium in SS2 is located inside the constrained region, reinforcing the idea that higher income risk leads households to demand more deposits and bonds, indirectly tightening constraints on firm financing. This endogenous shift in regime has important implications for both capital allocation and macroeconomic outcomes.

These results relate closely to [Bernanke and Gertler \(1989\)](#), where firm net worth governs borrowing constraints and amplifies real fluctuations. In contrast, this model generates financial frictions endogenously through the interaction between household saving behavior and firm financing choices. In [Bernanke and Gertler \(1989\)](#), the external finance premium is defined as the gap between external and internal funding costs, here a similar wedge arises between two external sources: bond financing and bank loans.²⁵ This wedge is zero when firms are unconstrained but becomes positive when borrowing frictions make one credit source relatively more expensive and it is determined endogenously by household savings and credit market conditions.

Thus, while in [Bernanke and Gertler \(1989\)](#) the external finance premium is driven by borrower net worth and affects access to all external funds, in my framework, the relative cost of external funds is shaped by shifts in the supply of liquid savings, which feed back into firm financing choices. Both mechanisms link financial frictions to real outcomes, but they operate through different margins, [Bernanke and Gertler \(1989\)](#) through the internal-external margin, and this paper through the bank-bond margin.

3.6.5 Liquidity premium

In aggregate, the total holdings of bonds and deposits will be positive if and only if the liquidity premium, as $LP_t \equiv r_t^b - r_t^d$, is positive.

Proposition 5. The liquidity premium is given by:

²⁵In [Bernanke and Gertler \(1989\)](#), this premium arises because lower net worth increases the risk of borrower default, raising the monitoring and agency costs borne by lenders.

- Under the unconstrained credit regime: $LP_t = \hat{c} > 0$
- Under the constrained credit regime:
 - If risk-free bond constraint binds: $LP_t = (1 - \kappa) \frac{K_{t+1}}{L_t^B} - (p_H \bar{R}_t^D - \hat{c})$
 - If risky bank loans constraint binds: $LP_t = \bar{R}_t^B - (p_H \phi \kappa \frac{K_{t+1}}{L_t^D} - \hat{c})$
 - If both constraints bind: $LP_t = (1 - \kappa) \frac{K_{t+1}}{L_t^B} - (p_H \phi \kappa \frac{K_{t+1}}{L_t^D} - \hat{c})$

Importantly, for sufficiently high monitoring costs $\hat{c}$, the liquidity premium remains positive regardless of the regime. This guarantees interior solutions in equilibrium, where firms use a mix of safe and risky debt. Moreover, the liquidity premium acts as a key pricing signal that shapes the allocation of funding across instruments and affects macro-financial transmission channels.

Besides, in equilibrium, when the bond constraint binds, an increase in aggregate bonds (L_t^B) generates two effects on the liquidity premium. First, the effective assets-to-bonds ratio, $(1 - \kappa) \frac{K_{t+1}}{L_t^B}$, declines as the denominator rises. Second, the net maximum cost of bank loans, $p_H \bar{R}_t^D - \hat{c}$, also falls because the upper bound on risky bank loans ($\bar{R}_t^D$) increases. Taken together, these two forces imply a negative overall effect on the liquidity premium.

Conversely, when the risky bank loans constraint binds, an increase in aggregate deposits (L_t^D) generates two effects on the liquidity premium. First, maximum cost of risk-free bonds, $\bar{R}_t^B$, increases. Second, the effective assets-to-bank loans ratio, $(p_H \phi \kappa \frac{K_{t+1}}{L_t^D})$, declines as the denominator rises. In combination, these two forces imply a positive overall effect on the liquidity premium.

4 Estimation

The dynamic problems 5 and 6 are solved numerically by discretizing the joint distribution $\Theta(d, b, z)$ over household states. In contrast, the dynamic problem 19 is solved at the representative firm level, since the linearity of firm policies implies that the distribution of net worth across firms does not affect aggregate outcomes. Consequently, the relevant demand functions for credit, defined in Equations 35 and 37, depend only on aggregate asset stock and interest rates. Further details of the algorithm are provided in the Appendix F.

Table 3 reports the parameters estimated for the period 1989-1995. I jointly estimate ten parameters; four related to households, two to banks, and four to firms, using the one-step Generalized Method of Moments (GMM) with the identity matrix as the weighting matrix. The parameters

are chosen to minimize the distance between a set of model-implied moments (e.g., household deposit shares, share of borrowers, firm’s leverage ratio, liquidity premium, etc.) and their empirical counterparts.

To achieve identification, some parameters are fixed following values commonly used in the literature. Specifically, the initial standard deviation of the innovation to log-income ($\sigma = 0.4$), the persistence of the income process ($\rho = 0.9$), the coefficient of relative risk aversion ($\gamma = 2.45$)²⁶, the capital share ($\alpha = 0.4$)²⁷, and the labor income tax ($\tau^g = 0.3$) are held fixed at values commonly used in the literature.²⁸ After estimating the model, I adjust σ in a manner consistent with the observed rise in income inequality, explicitly targeting the increase in the P90/P10 income ratio over the last 30 years. In this exercise, I assume a depreciation rate of zero to identify the monitoring effectiveness parameter (ϕ) and the partial loss of asset stock parameter (κ). Under this assumption, the firm’s asset stock should be interpreted broadly to encompass any factor of production that must be financed in advance, such as intangible, physical, or organizational capital. This generalization allows the model to capture a wide range of investment needs while focusing on how financial frictions shape firm behavior.

Table 4 reports the eleven targeted moments that discipline the model using key household and corporate financial variables. Since the objective of the estimation is to evaluate how rising income inequality shapes the corporate debt structure, I combine information from two complementary data sources. The average deposits-to-bonds ratio is taken from the 1989 SCF, which provides a detailed benchmark of the wealth distribution across households and thus captures how balance sheets are allocated between liquid and illiquid assets at the start of the period. In contrast, the cross-sectional deposit shares are derived from the 1989-1995 DFA, which provides consistent annual information on the distribution of disposable income, allowing me to track how income inequality maps into savings composition in the early 1990s. Using both sources ensures that the estimation is anchored in the wealth heterogeneity observed in the SCF while also capturing the income-based reallocation patterns documented in the DFA.

In addition to the deposits-to-bonds ratio and the cross-sectional deposit shares, I also target the share of borrowers. This moment is crucial for identification, as it helps pin down the scale parameter of adjustment costs (σ_x) and prevents the model from allocating an unrealistically large share of aggregate resources to deposits that would otherwise be transformed into risky bank loans

²⁶Hall (2016) implements the model using $\gamma \in \{2.0, 2.5\}$.

²⁷Bergholt et al. (2022) document a structural decline in the U.S. labor share over the past two decades. For their Monte Carlo exercises, they adopt a median labor share of 0.6, which corresponds to a capital share of 0.4.

²⁸Aiyagari (1994); Bergholt et al. (2022); Hall (2016); Bayer et al. (2019).

Table 3: Estimated Parameters

Parameter	Value	Description	Target
<i>Households</i>			
σ	0.4	Innovation of productivity	Aiyagari (1994)
ρ	0.9	Persistence of productivity	Aiyagari (1994)
γ	2.45	Relative risk aversion	Hall (2016)
β	0.945	Discount factor	Table 4
$\hat{r}$	5.7%	Penalty rate	Table 4
μ_χ	562.63	Mean adjustment cost	Table 4
σ_χ	194.13	Scale adjustment cost	Table 4
<i>Banks</i>			
$\hat{c}$	0.06	Monitoring cost	Table 4
ϕ	0.88	Monitoring effectiveness	Table 4
<i>Firms</i>			
α	0.4	Capital share	Bergholt et al. (2022)
p_H	0.996	Probability partial loss	Table 4
κ	0.70	Partial loss	Table 4
A	4.04	Technology	Table 4
τ	7.11%	Exit probability	Table 4
<i>Government</i>			
τ^g	0.3	Labor income tax	Bayer et al. (2019)

in general equilibrium. By disciplining the extent of borrowing activity, it ensures that the debt market is not matched solely through savings behavior.

To discipline the model, I estimate the discount factor, the parameters governing the adjustment cost distribution, and the penalty rate of borrowing ($\hat{r}$) to match key household moments: the aggregate deposit intensity ($\frac{D}{B}$), cross-sectional deposit shares, and the share of borrowers. On the firm and banking side, I estimate the parameters governing firm exit and idiosyncratic shocks to replicate the relevance of external funding ($\frac{D+B}{N}$), the ratio of bank loans to assets, and bond returns. Finally, parameters associated with banks' monitoring technology are calibrated to capture collateral values and the liquidity premium.

The first important result is that while the model overstates the aggregate deposit-to-bond ratio relative to the data (Model: 2.13 vs. Data: 1.02), this reflects a trade-off in the estimation that prioritizes matching the cross-sectional deposit shares and the liquidity premium. Regarding the deposit share across the income distribution, the model predicts slightly higher values, particularly at the bottom and top, but it nevertheless reproduces the monotonic pattern observed in the DFA. The model also accounts for nearly half of the total share of borrowers. Because all household

Table 4: Targeted moments

Targets	Model	Data	Source	Parameter
Deposits / Bonds $\frac{D}{B}$	2.13	1.02	SCF (1989)	Discount factor
Deposit share (1st quintile)	88.4	86.6	DFA (1989-1995)	Scale adj.costs
Deposit share (3rd quintile)	74.2	80.1	DFA (1989-1995)	Scale adj.costs
Deposit share (Top 1%)	30.5	28.3	DFA (1989-1995)	Mean adj.costs
Share of borrowers (%)	4.6	10.0	SCF (1989)	Penalty rate
Ext. funding / net worth $\frac{(D+B)}{N}$	2.18	1.97	Z.1 FA (1990-1993)	Exit probability
Bank loans / Assets $\frac{D}{K}$	0.47	0.21	Z.1 FA (1990-1993)	Prob. partial loss
Collateral for banks	0.57	0.59	Gupta et al. (2022)	Monit. effectiveness
Assets / Sales $\frac{K}{Y}$	5.04	4.54	FRED (1992-1993)	Technology
Bond return	7.65%	7.27%	FRED (1989-1993)	Partial loss
Liquidity premium	5.88%	5.46%	FRED (1989-1993)	Monitoring cost

savings are assumed to be intermediated into firm credit, the aggregate ratio of deposits to bonds directly maps into the firm-level corporate bond structure, defined as $\frac{B}{B+D}$. The model produces a bond share of 32.0%, which corresponds to 47% of the empirical corporate bond structure recorded in the data.²⁹

Importantly, the model does not incorporate international investors or foreign capital inflows, which in reality provide a significant share of corporate bond financing. Despite this limitation, the model captures a substantial portion of the observed corporate bond structure, underscoring its ability to replicate key financial ratios using only domestic household savings and intermediation mechanisms.

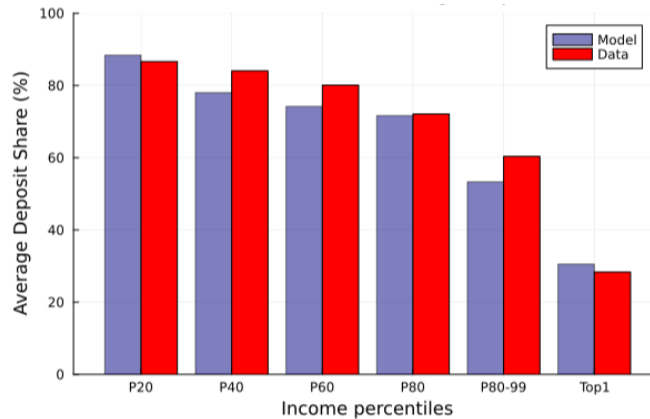
A second key result is that the model predicts a higher reliance on external funding than reported in the Z.1 Financial Accounts for the period 1990-1995 (Model: 2.18 vs. Data: 1.97). This gap reflects the model’s simplifying assumption that all household savings, particularly deposits, are fully channeled into firms’ credit. In reality, banks hold reserves, and by abstracting from these reserves, the model overstates the supply of credit available to firms. Despite these simplifying assumptions, the model closely matches several other corporate firm financing ratios, including the ratio of bank loans to asset stock, the share of collateralizable assets held by banks measured by the upper-bound of bank loans-to-assets ($\frac{\phi\kappa}{RD}$), and the asset-to-sales ratio. Finally, the model almost matches the real return of high-quality corporate bonds and the liquidity premium implied

²⁹According to the Z.1 Financial Accounts, U.S. corporations reported an average bond share of 67.8% between 1990 and 1993. The model reproduces a value of 32.0%, capturing approximately 47% of the observed structure.

by the data for the period 1989-1993, as reported by FRED.^{30,31}

Figure 4 presents the average deposit share by income percentile using DFA data from 1989 to 1995, along with the model-implied estimates. Income is defined as the sum of labor and capital income. The model closely replicates the downward-sloping pattern observed in the data, capturing the fact that lower-income households hold a higher share of deposits relative to bonds compared to those at the top of the distribution.

Figure 4: Deposit shares across income percentiles (Model vs. Data)



Notes: This figure displays the estimated shares of deposits across income percentiles compared to data from the Distributional Financial Accounts for 1989-1995. Income in the model is computed as the sum of labor income, capital income (returns on deposits and bonds, annualized), and dividends. The estimated relationships are obtained using a local linear regression with a Gaussian kernel and a bandwidth of 0.025.

Finally, Figure A12 presents two Lorenz curves: the blue line reflects the distribution implied by the model, while the red line represents the Lorenz curve constructed from 1989 SCF data on liquid wealth (defined as deposits and bond holdings). Consistent with the visual comparison, the Gini coefficient implied by the model indicates lower liquid wealth inequality than the empirical Gini coefficient.

³⁰The real return of corporate bonds is calculated as the average of the HQMCB1YR and HQMCB2YR series, adjusted for inflation using the CPIAUCNS series. All data are sourced from the FRED portal.

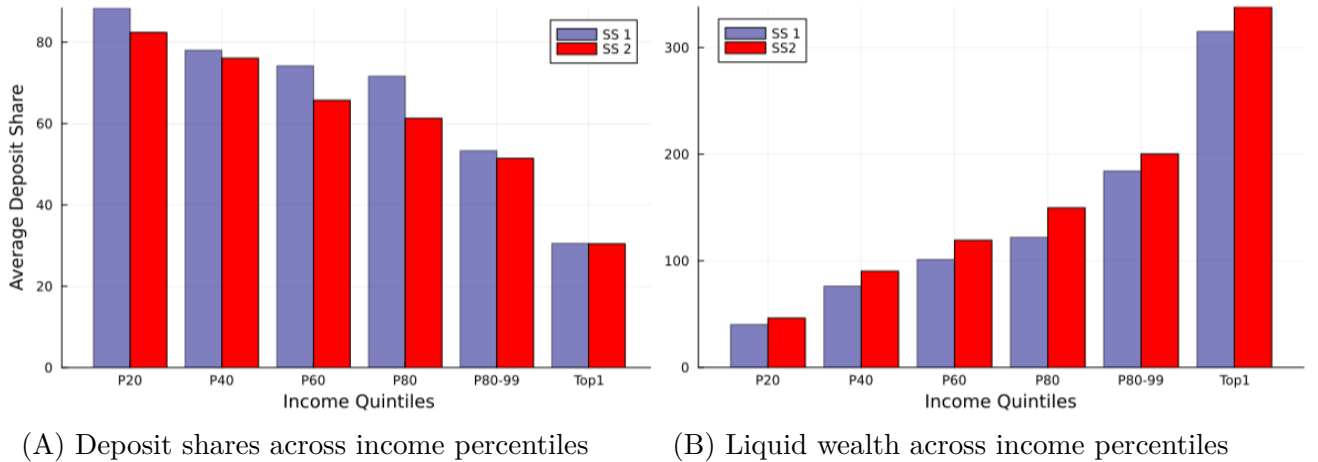
³¹The liquidity premium is computed as the difference between the real return on corporate bonds and the 3-month commercial paper rate for AA-rated firms, using the series CP3M and CPN3M. Since CP3M was discontinued in 1997, I have complemented it with CPN3M from that point onward. All data was adjusted by inflation using the CPIAUCNS series.

5 Quantitative results

In this section, I examine how the rise in income inequality in the United States over the past three decades affects the model's key outcomes. The analysis interprets this increase in inequality as driven solely by greater dispersion in idiosyncratic labor productivity shocks, consistent with mechanisms documented in the literature and discussed in Subsection 2.1 regarding the persistent widening of U.S. wage dispersion.

Based on the CPS data during the periods 1985-1990 and 2015-2019, income inequality, measured by the P90/P10 income ratio, increased by 24.3%, from 10.06 to 12.50, which I interpret as reflecting an 8.4% rise in labor income risk.³² As households face greater income risk, they adjust their savings behavior accordingly. Figure 5B shows that households consistently accumulate more savings, while reducing the share held in deposits across the income distribution. The decline in deposit shares is most pronounced for households in the bottom and middle of the distribution (Figure 5A).

Figure 5: Estimated deposit shares and liquid wealth (Baseline - SS1 vs. Counterfactual - SS2)



Notes: Figure A displays the estimated shares of deposits across income percentiles before and after the increase in labor income risk. Figure B displays the estimated liquid wealth (deposits plus bond holdings) across income percentiles before and after the increase in labor income risk.

Consequently, the aggregate deposits-to-bonds ratio declines by 0.41 points in levels (from 2.13 to 1.72), driven mainly by households in the middle of the income distribution. Deposit shares fall across the distribution, with the sharpest reductions observed in the middle quintile (down 8.5

³²I assume that the increase in the P90/P10 ratio is exclusively driven by rising labor income dispersion.

percentage points, from 74.2% to 65.7%) and the bottom quintile (down 6.0 percentage points, from 88.4% to 82.4%). Households in the top 1% reduce their deposit share only marginally, from 30.5% to 30.4%, reflecting their already solid reliance on bonds. Overall, the model implies a consistent shift toward bond holdings across the distribution, with wealthier households continuing to hold a substantial share in bonds. On the firm side, the model predicts a 15.0% increase in the corporate bond share, from 32.0% to 36.8%, as firms rely more heavily on bond financing. This response is qualitatively consistent with the first empirical pattern documented in the motivating evidence of this paper. Quantitatively, it corresponds to an elasticity of 1.79, meaning that a 1% increase in labor income risk generates a 1.79% rise in bond reliance.

Table 5: Selected moments: Baseline and counterfactual stationary equilibrium

Moments	Baseline SS 1	Counterfactual SS 2	Change
Std. dev. of productivity	0.40	0.43	8.4%
Implied corp. bond structure $\frac{(B)}{B+D}$	32.0%	36.8%	15.0%
Deposits / Bonds $\frac{D}{B}$	2.13	1.72	-0.41
Deposit share (1st quintile)	88.4	82.4	-5.98
Deposit share (3rd quintile)	74.2	65.7	-8.47
Deposit share (Top 1)	30.5	30.4	-0.1
Gini coefficient (Liquid wealth)	0.42	0.40	-0.02
Share of borrowers (%)	4.6	4.2	-0.4
Consumption Inequality P90/P10	3.77	3.76	-0.01
Ext. funding / net worth $\frac{(D+B)}{N}$	2.18	3.30	1.12
Assets / Sales $\frac{K}{Y}$	5.04	5.12	0.08
Bond return	7.65%	6.29%	-1.37
Liquidity premium	5.88%	4.16%	-1.73

Regarding liquid wealth inequality, the model predicts a modest decline in the Gini coefficient, by 0.02 points (from 0.42 to 0.40). This reduction is driven by changes in saving behavior: households in the middle and lower parts of the income distribution substantially reduce their reliance on deposits and increase their holdings of bonds, thereby raising their liquid wealth. Nevertheless, wealthier households continue to accumulate a disproportionate share of total liquid wealth, so the overall decline in inequality remains limited. Consistent with this result, the model also predicts a decline in the share of borrowers, from 4.6% to 4.2%, indicating that fewer households are constrained and more can self-finance through savings. As income risk rises and borrowing becomes more costly, households reduce their reliance on consumer credit.

Additionally, the model predicts an increase in both corporate leverage and the assets-to-sales

ratio. Specifically, the ratio of external funding to firms' net worth rises by 1.12 points, from 2.18 to 3.30, reflecting greater reliance on external finance. At the same time, the assets-to-sales ratio increases by 0.08 points, from 5.04 to 5.12. These changes are driven by a higher level of liquid savings among households, particularly as bonds become increasingly attractive relative to bank loans, thereby reducing their cost and encouraging firms to tilt their debt structure toward bond financing.

Regarding the liquidity premium, the model predicts a decline of 1.73 percentage points (from 5.88% to 4.16%), which is qualitatively consistent with the third empirical pattern highlighted in the motivating evidence of this paper. A regime shift on the firm side of the model drives this change. As aggregate bond holdings increase in response to higher income risk, the supply of funds in the bond market expands to the point where the risk-free bond constraint becomes binding. This transition leads to a decline in the bond interest rate as bond credit demand becomes inelastic in the constrained regime, unlike in the unconstrained regime, where the demand curve is perfectly elastic.

Table 6: Returns and wages in equilibrium

Factor return	Baseline SS 1	Counterfactual SS 2	Change %
Bonds return (r^b)	7.65%	6.29%	-17.8%
Deposits return (r^d)	1.77%	2.13%	20.3%
Wages (W)	18.05	18.25	1.1%

Table 6 reports the equilibrium outcomes under two steady states: a baseline and a counterfactual with higher income inequality. The increase in inequality is reflected in a rise in the standard deviation of idiosyncratic labor productivity, from 0.40 to 0.43. This shock amplifies labor income risk and deepens the heterogeneity across households' after-tax labor income and wealth accumulation paths.

The general equilibrium response includes adjustments in key prices. The return on bonds declines from 7.65% to 6.29% (a 17.8% decrease), reflecting a greater demand for illiquid bonds. Meanwhile, the deposit rate increases from 1.77% to 2.13% (a 20.3% rise), driven by the firms' higher cost of bank loans compared to bonds. Also, wages experience a modest 1.1% increase, due to a rise in firms' stock of assets.

Table 7 reports the equilibrium aggregates across the two stationary equilibria. Aggregate consumption remains essentially unchanged, falling by only 0.2%. In contrast, total savings rise by 15.0%, driven primarily by a 32.3% increase in bond holdings alongside a 6.9% rise in deposits.

On the firm side, net worth declines sharply by 24.0%, reflecting firms’ substitution away from internal financing toward cheaper external debt as bond returns fall (Table 6). This shift lowers the cost of market borrowing and reduces reliance on retained earnings. At the macroeconomic level, total assets expand by 2.7% and output rises by 1.1%, consistent with higher capital accumulation facilitated by more favorable external financing conditions.

Table 7: Aggregates in equilibrium

Aggregate variables	Baseline SS 1	Counterfactual SS 2	Change %
Consumption	20.01	19.97	-0.2%
Deposits	70.47	73.34	6.9%
Bonds	33.15	43.84	32.3%
Total savings	103.62	119.18	15.0%
Output	29.97	30.29	1.1%
Firms’ net worth	47.50	36.10	-24.0%
Total assets	151.11	155.21	2.7%

5.1 Effect of Financial Asset Returns, Wages, and Income Inequality

In this section, I decompose the total change in firms’ bond reliance into two components: the partial equilibrium effect, which captures the impact of higher household income risk holding prices fixed, and the general equilibrium effect, which incorporates the endogenous adjustment of prices. Initially, firms’ reliance on bond financing is 32.0%. When income risk rises but prices remain unchanged, bond reliance increases modestly to 33.8%. Allowing prices to adjust further amplifies this effect, raising bond reliance to 36.8%. This decomposition indicates that roughly 37.5% of the total change is driven by higher income risk itself, while 62.5% arises from general equilibrium feedback through prices and portfolio reallocation.

To deeply understand the effect of only higher income risk, Figure 6 provides a decomposition of the total change in households’ asset holdings across income percentiles. Figure 6A displays changes in bond holdings, while Figure 6B shows changes in deposit holdings. Each bar separates the contribution of four channels: the pure effect of higher inequality (blue), the joint effect of higher inequality and bond returns (red), the joint effect of higher inequality and deposit returns (green), and the joint effect of higher inequality and wages (orange). The black line plots the total joint effect, summarizing the combined impact across channels.

Figure 6A shows that bond holdings rise most strongly in the upper tail, particularly for households

in the fourth quintile and the top 1%. Although the sharp decline in bond returns (from 7.65% to 6.29%) dampens the incentive to hold bonds, top-earners households continue to expand their bond positions. This reflects their relatively smaller portfolio adjustment frictions, which make reallocating precautionary savings into bonds more attractive than into deposits. On the other hand, Figure 6B highlights a different pattern for deposits. The largest increase in deposits is concentrated among households in the second quintile, while the top 1% also expand their deposit holdings, though mainly as a byproduct of the interaction between higher inequality and falling bond returns. By contrast, the households in the third and fourth quintiles reduce their deposit exposure, substituting into bonds when returns are favorable (red bars).

Moreover, households at the top of the distribution benefit the most from the increase in labor productivity dispersion and the accompanying rise in wages. Although the average wage gain is modest (around 1.1%), the higher variance in productivity enables top-income households to strengthen their wealth accumulation, primarily through greater bond holdings. In this context, rising inequality amplifies precautionary saving motives among high-income earners, encouraging a portfolio shift toward bonds. However, the general equilibrium decline in bond returns partially offsets this effect, reducing the incentive to hold bonds, particularly for the top 1% of households, whose portfolios are already heavily concentrated in these assets.

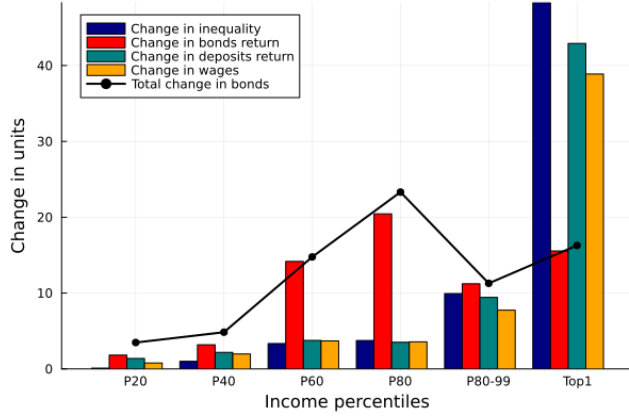
Lastly, Figure 6D shows how liquid wealth composition adjust across income percentiles when inequality rises in partial and general equilibrium. The blue bars indicate that higher inequality alone reduces deposit shares most among top-income households, reflecting their lower exposure to adjustment costs. In general equilibrium, middle-income households shift part of their deposits toward bonds despite declining bond returns, as their higher income risk raises the marginal value of holding bonds. By contrast, top households who are already bond-rich, face a lower marginal value of additional bond holdings, leading to minimal changes in deposit shares. Together, these results highlight heterogeneous portfolio responses to rising inequality. While households across the distribution reduce their deposit shares, upper-middle households do so the most, generating redistributive implications that will be discussed in the next subsection.

5.2 Distributional Effects on Liquid Wealth Inequality

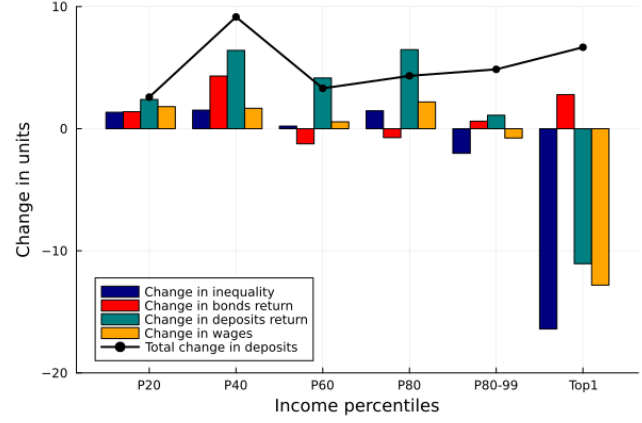
Figure 6C illustrates the decomposition of changes in total liquid wealth across income percentiles. This decomposition highlights the heterogeneous impact of inequality and price adjustments across the income distribution. The figure reveals that all groups experience an increase in liquid wealth in response to rising income inequality, but the relative contributions vary.

Figure 6: Decomposition of the total effect of higher income inequality

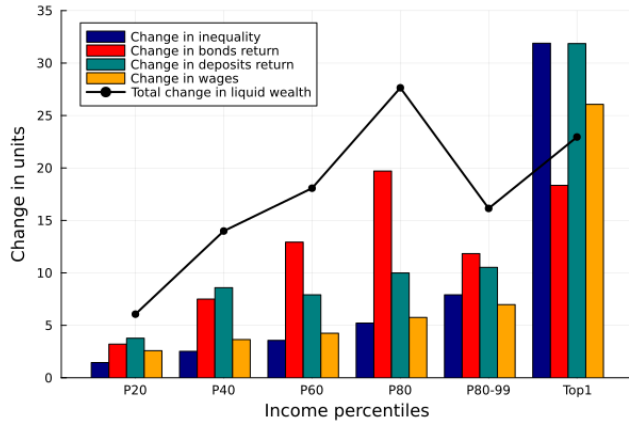
(A) Change in bonds



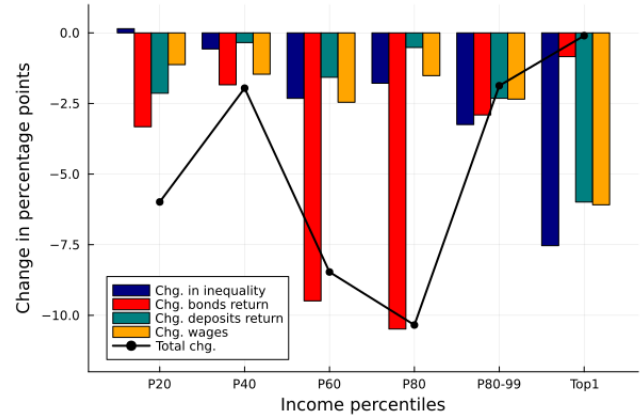
(B) Change in deposits



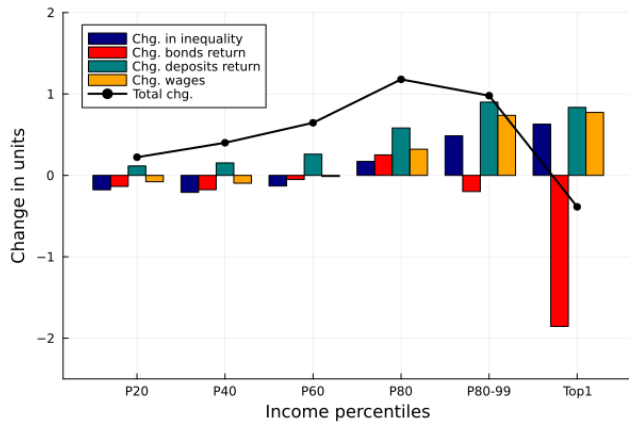
(C) Change in liquid wealth



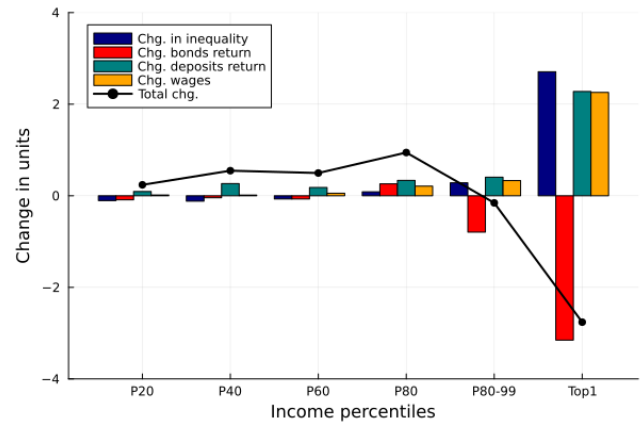
(D) Change in deposit shares



(E) Change in total income



(F) Change in consumption



Notes: Figures (A) - (F) show how changes in bond and deposit holdings vary across wealth percentiles in response to higher income inequality. Blue bars isolate the effect of higher inequality; red, green, and orange bars reflect the joint effect of higher inequality and changes in bond returns, deposit returns, and wages, respectively. The black line depicts the combined effect of all three channels.

Households in the upper-middle of the distribution see the most significant absolute gains, primarily driven by the joint effects of higher income inequality and changes in financial asset returns. While the top 1% continues to accumulate substantial liquid wealth, the increase is less disproportionate compared to upper-middle households.

This more evenly distributed growth in liquid assets helps explain the model's prediction of a decline in the Gini coefficient for liquid wealth, from 0.42 to 0.40. In particular, the stronger response among middle-income households reflects the lower portfolio adjustment frictions relative to households at the bottom. As these households reallocate toward bonds, inequality in liquid wealth holdings compresses, despite rising income inequality, highlighting the redistributive role of return dynamics in general equilibrium. However, a more specific comparison of liquid wealth across the income distribution reveals that inequality is still increasing when comparing households in the fourth quintile with lower income groups. For example, the P80/P20, P80/P40, and P80/P60 ratios rise 6.61% (from 3.03 to 3.23), 3.64% (from 1.60 to 1.66), and 4.08% (from 1.21 to 1.26), respectively, signaling an increase in upper-middle-to-bottom wealth inequality.

Conversely, inequality narrows within the top and the lower income groups. For instance, the top 1%/P20 and top 1%/P80 ratios decline by 6.75% (from 7.82 to 7.29) and 12.53% (from 2.58 to 2.26), respectively. This pattern suggests that while most of the income distribution experiences a surge in liquid wealth inequality, households in the lower percentiles increase their liquid wealth proportionally more than those in the top 1%, thereby reducing the relative gap between the very top and the rest of the distribution.

5.3 Distributional Effects on Income Inequality

Figure 6E shows that rising income inequality reshapes the income distribution both directly and through its interaction with financial returns and general equilibrium feedback. The partial equilibrium effect (blue bars) follows a monotonic pattern: income falls at the bottom and rises toward the top. Once price adjustments are incorporated, this pattern weakens since general equilibrium effects alter returns and dampen the direct impact of income dispersion. For lower-middle households, higher deposit returns yield modest gains, upper-middle households benefit the most, while the top 1% face losses due to falling bond returns. The hump-shaped total effect (black line) highlights how return dynamics redistribute income gains away from the top 1% toward the rest.

Comparing total income across percentiles shows that inequality continues to rise between upper-

middle and lower-income groups. The P80/P20, P80/P40, and P80/P60 ratios increase by 1.12% (from 4.21 to 4.26), 1.26% (from 2.27 to 2.30), and 0.95% (from 1.51 to 1.53), respectively, indicating greater upper-middle-to-bottom inequality. By contrast, the top 1%/P20 and top 1%/P80 ratios fall by 5.04% (from 12.20 to 11.58) and 6.10% (from 2.90 to 2.7), showing that the very top loses ground relative to both lower and upper-middle groups due to their exposure to falling bond returns. Overall, rising inequality is driven mainly by upper-middle households, while the top 1% experiences relative losses.

5.4 Distributional Effects on Consumption Inequality

Figure 6F shows that the partial equilibrium effect (blue bars) follows a monotonic pattern as in the case of income. Adding general equilibrium effect, the overall effect is positive for most households in the lower and middle percentiles, reflecting both higher deposit returns. The gains are relatively modest for the first, second, and third quintiles but peak at the fourth quintile, where consumption rises the most. By contrast, the top 1% experience a sharp decline in consumption, primarily driven by their exposure to falling bond returns. The red bar in Panel Top 1% highlights how the drop in r^b disproportionately reduces disposable resources for the wealthiest households, leading to a steep fall in their consumption due to lower wealth effect.

This pattern mirrors the hump-shaped profile in income but with stronger distributional effects. Consumption inequality rises between upper-middle and lower-income groups, as the P80/P20, P80/P40, and P80/P60 ratios increase by 2.02% (from 2.37 to 2.42), 0.45% (from 1.57 to 1.58), and 1.77% (from 1.23 to 1.26), respectively. By contrast, the top 1%/P20 and top 1%/P80 ratios fall by 9.26% (from 5.63 to 5.11) and 11.06% (from 2.37 to 2.11), indicating that the very top loses ground relative to both the bottom and upper-middle. Overall, rising consumption inequality is driven mainly by upper-middle households, while top earners experience relative declines.

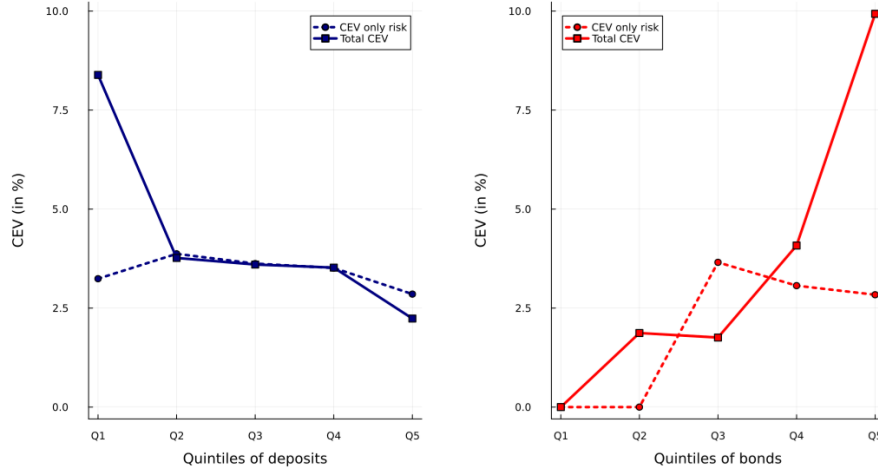
The relationship between income and consumption inequality is a topic far from reaching consensus within the profession. These results, which exclude the households in the top 1%, complement the evidence in [Aguiar and Bils \(2015\)](#); [Attanasio et al. \(2012\)](#), who show that consumption inequality closely mirrors income inequality when comparing high- and low-income households across different categories of goods. However, [Meyer and Sullivan \(2023\)](#) and [Heathcote et al. \(2010a\)](#) conclude that the rise in overall consumption inequality was small.

Discussion about welfare: To quantify the welfare implications of rising income inequality, I compute the consumption equivalent variation (CEV) between the two stationary equilibria of the

model.³³ The CEV measures the permanent percentage change in consumption that households would require in the high-inequality economy (SS2) to be as well off as in the baseline economy (SS1). In other words, the following calculations answer the question: What compensation would households, after the economy has re-sorted in SS2, need to be as well off as in SS1?

On average, households would need to be compensated by 4.04% of their consumption to offset the welfare losses associated with higher income dispersion. In a complementary exercise, I compute the CEV while holding all prices constant at their initial stationary equilibrium levels and only increasing the dispersion of productivity shocks. Under this partial equilibrium scenario, households would require a smaller compensation of 3.35% to remain as well off as before inequality rose.

Figure 7: Welfare analysis: Partial vs. Total compensation



Notes: This figure reports the average consumption-equivalent variation (CEV) by quintiles of the bond (left panel) and deposit (right panel) distributions. The total CEV measures the additional consumption (in %) that households in the high-inequality economy would require to be as well off as in the low-inequality economy, accounting for changes in equilibrium prices. The partial CEV measures the additional consumption (in %) households would need under higher income dispersion but with prices fixed at their low-inequality stationary equilibrium levels. In this latter exercise, the average CEV was calculated using the distribution in the high-inequality economy with fixed prices.

Figure 7 decomposes the welfare effects of rising income inequality into partial and total compensations, measured as the consumption-equivalent variation (CEV) across quintiles of deposits and bonds. The dashed lines represent the partial equilibrium effects, computed by increasing the dispersion of idiosyncratic productivity shocks while holding all prices fixed, whereas the solid

³³The average CEV was calculated in two steps. First, I found the value λ_{CEV} that solves $V_{SS1}(d, b, z) = V_{SS2}(\lambda_{CEV}, d, b, z)$. Second, the average CEV was computed as $\bar{\lambda}_{CEV} = \Phi_{SS2}\lambda_{CEV}$.

lines represent the total general equilibrium effects, which also incorporate changes in wages and asset returns. A key takeaway from this comparison is that partial compensations are more evenly distributed across households within the deposits and bonds distributions, indicating that general equilibrium feedback amplify welfare heterogeneity.

Among deposit-rich households (left panel), welfare losses are largest at the bottom of the deposit distribution. These households rely heavily on deposits for consumption smoothing, so when income risk rises and borrowing becomes more costly, they face tighter liquidity and stronger precautionary motives, requiring nearly an 8% compensation in consumption to remain indifferent. In contrast, wealthier deposit holders experience smaller losses. The modest gap between partial and total CEVs for deposit-rich households suggests that general equilibrium effects, particularly higher deposit returns, do little to offset the welfare costs of greater income volatility.

Bond-rich households (right panel) exhibit the opposite pattern. Welfare losses rise sharply toward the top of the bond distribution. As inequality increases, falling bond returns reduce the payoff on their preferred asset, leading to the largest welfare declines. While partial equilibrium effects (dashed line) show moderate losses, the total CEV (solid line) reveals that price adjustments, especially lower bond rates, substantially amplify the welfare costs for bond savers.

5.5 Comparing Predicted and Observed Changes

Table 8 summarizes changes in key moments from both the data and the model. The model successfully replicates the decline in the aggregate deposits-to-bonds ratio between the 1989 and 2016 SCF and predicts a substantial rise in corporate bond reliance, accounting for 59.5% of the increase observed in the data. It also captures broad distributional patterns in savings behavior. For example, it accounts for 70.7% of the observed decline in the deposit share of the third quintile and 36.8% for the first quintile. By contrast, the model underpredicts changes at the very top, yielding only a 0.1 percentage point decline in the deposit share of the Top 1%, whereas the data show an increase of 15.3 percentage points. This discrepancy reflects the model’s narrow focus on income risk as the sole driver of savings composition. While rising income risk generates a shift toward bonds for lower and middle-income households, through higher equilibrium wages and a lower liquidity premium, it does not capture other mechanisms that led top-income households to raise their deposit holdings.

One potential explanation is that the DFA may not fully capture households’ liquid saving behavior at the very top of the distribution. As Figure A4 shows, liquid wealth in the top 1% may partly

reflect measurement or compositional issues rather than direct household choices. For instance, top households often delegate portfolio management to mutual funds or other intermediaries (such as private "family offices"). Despite these limitations, I target this moment because it remains the best available benchmark for assessing the composition of liquid wealth among U.S. households.

Table 8: Change in moments (Model vs. Data)

Moments	Change-Model	Change-Data	Source
Implied corp. bond structure $\frac{(B)}{B+D}$	15.0%	25.2%	Z.1 FA (1990-2019)
Deposits / Bonds $\frac{D}{B}$	-0.41	-0.40	SCF (1989-2016)
Deposit share (1st quintile)	-5.98	-16.28	DFA (1989-2016)
Deposit share (3rd quintile)	-8.47	-11.98	DFA (1989-2016)
Deposit share (Top 1)	-0.1	15.31	DFA (1989-2016)
Share of borrowers (%)	-0.39	-0.30	SCF (1989)
Gini coefficient (Liquid wealth)	-0.02	0.11	SCF (1989-2016)
Bond return	-1.37	-5.70	FRED (1989-2019)
Liquidity premium	-1.73	-2.99	FRED (1989-2019)
Ext. funding / net worth $\frac{(D+B)}{N}$	1.12	3.46	Z.1 FA (1989-2016)
Assets / Sales $\frac{K}{Y}$	0.08	0.37	FRED (1992-2019)

Moreover, while the model predicts a slight decline in liquid wealth inequality, driven by upper-middle households accumulating more liquid savings and thus compressing the distribution, the data show an increase of 0.11 in the Gini coefficient, based on SCF estimates. This discrepancy may reflect the model's abstraction from forces such as the relaxation of borrowing constraints, which enabled households at the bottom to dissave (Mian et al., 2020), and likely shaped U.S. households' portfolio choices between 1989 and 2019.

Finally, the model closely matches the decline in the liquidity premium, accounting for 57.8% of the drop observed in the data. Overall, these results highlight the model's ability to replicate key structural patterns in household behavior and firm financing in response to rising labor income risk.

6 Conclusions

This paper formalizes a novel mechanism linking income inequality, corporate debt structure, and the liquidity premium. It delivers three main contributions. First, this paper offers a structural explanation for changes in household financial intermediation and the evolution of the corporate

debt structure. These results underscore the model’s ability to link household portfolios to aggregate credit market outcomes, even under simplified assumptions, providing a robust foundation for analyzing structural transformations in financial markets.

Second, the model delivers a strong quantitative fit to key macro-financial patterns observed in the U.S. between 1989 and 2019. By increasing the dispersion of labor productivity shocks to match the rise in the P90/P10 income ratio, it explains 59.5% of the observed shift in firms’ financing structure. It also captures the heterogeneous reallocation of household savings across the income distribution, including the decline in deposit intensity among top households and the relative shift toward bond holdings, especially at the middle. Moreover, it partially matches the empirical reduction in the liquidity premium, accounting for 57.9% of the observed drop.

Third, the model shows that rising income inequality generates redistributive effects on household portfolios. Higher labor income risk raises liquid wealth overall but compresses it between the top 1% and lower-income households. Contrary to standard intuition, the liquid-wealth Gini slightly declines (from 0.42 to 0.40) as middle-income households, facing lower adjustment frictions relatively to lower income households, shift more toward bonds. This reallocation reduces financial inequality despite greater income risk, highlighting the redistributive role of return dynamics in general equilibrium. At the same time, both income and consumption inequality widen between the upper and lower ends of the distribution: the P80/P20 ratio rises from 4.21 to 4.26 for income and from 2.37 to 2.42 for consumption, reflecting larger gains for upper households and tighter borrowing and adjustment constraints at the bottom.

Finally, this framework offers a foundation for analyzing the distributional and macro-financial effects of fiscal and financial policies. In particular, the model can be extended to study targeted tax policies, such as progressive capital income taxation or redistributive transfers, and their impact on household saving composition, asset prices, and credit market dynamics. Given its rich heterogeneity and general equilibrium structure, it is well-suited to evaluate how taxes that alter households’ disposable income or relax borrowing constraints influence both inequality and firms’ financing choices. Such extensions could shed light on the trade-offs faced by policymakers aiming to reduce inequality and help clarify the role of fiscal tools in economies experiencing persistent income inequality.

References

- Aguiar, M. and Bils, M. (2015). Has consumption inequality mirrored income inequality? *The American economic review*, 105(9):2725–2756.
- Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly journal of economics*, 109(3):659–684.
- Allen, K., Saha, P., Whitley, M., and Winters, D. (2023). Money market reforms: the effect on the commercial paper market. *Journal of banking & finance*, 154:106947–.
- Attanasio, O., Hurst, E., and Pistaferri, L. (2012). The evolution of income, consumption, and leisure inequality in the us, 1980-2010. *NBER Working Paper Series*, pages 17982–.
- Auclert, A., Bardoczy, B., Rognlie, M., and Straub, L. (2021). Using the sequence-space jacobian to solve and estimate heterogeneous-agent models. *Econometrica*, 89(5):2375–2408.
- Auclert, A., of Economic Research., N. B., and Rognlie, M. (2018). *Inequality and Aggregate Demand / Adrien Auclert, Matthew Rognlie*. NBER working paper series no. w24280. National Bureau of Economic Research, Cambridge, Mass.
- Auclert, A., Rognlie, M., and Straub, L. (2024). The intertemporal keynesian cross. *The Journal of political economy*, 132(12):4068–4121.
- Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–2168.
- Autor, D. H., Katz, L. F., and Kearney, M. S. (2006). The polarization of the u.s. labor market. *The American economic review*, 96(2):189–194.
- Bayer, C., Luetticke, R., Pham-Dao, L., and Tjaden, V. (2019). Precautionary savings, illiquid assets, and the aggregate consequences of shocks to household income risk. *Econometrica*, 87(1):255–290.
- Becker, B. and Ivashina, V. (2014). Cyclical credit supply: Firm level evidence. *Journal of monetary economics*, 62:76–93.
- Bergholt, D., Furlanetto, F., and Maffei-Faccioli, N. (2022). The decline of the labor share: New empirical evidence. *American economic journal. Macroeconomics*, 14(3):163–198.

- Bernanke, B. and Gertler, M. (1989). Agency costs, net worth, and business fluctuations. *The American economic review*, 79(1):14–31.
- Bernanke, B., of Economic Research., N. B., Gertler, M., and Gilchrist, S. (1998). *The Financial Accelerator in a Quantitative Business Cycle Framework / Ben Bernanke, Mark Gertler, Simon Gilchrist*. NBER working paper series no. w6455. National Bureau of Economic Research, Cambridge, Mass.
- Buchak, G., of Economic Research., N. B., Matvos, G., Piskorski, T., and Seru, A. (2024). *The Secular Decline of Bank Balance Sheet Lending / Greg Buchak, Gregor Matvos, Tomasz Piskorski, Amit Seru*. NBER working paper series no. w32176. National Bureau of Economic Research, Cambridge, Mass.
- Cao, D. and Luo, W. (2017). Persistent heterogeneous returns and top end wealth inequality. *Review of economic dynamics*, 26:301–326.
- Card, D. (1998). Falling union membership and rising wage inequality: What’s the connection? *NBER Working Paper Series*, pages 6520–.
- Coibion, O., Gorodnichenko, Y., Kueng, L., and Silvia, J. (2017). Innocent bystanders? monetary policy and inequality. *Journal of monetary economics*, 88:70–89.
- Crouzet, N. (2021). Credit disintermediation and monetary policy. *IMF economic review*, 69(1):23–89.
- Doerr, S., Drechsel, T., and Lee, D. (2024). Income inequality and job creation. *NBER Working Paper Series*.
- Doidge, C., Karolyi, G. A., and Stulz, R. M. (2013). The u.s. left behind? financial globalization and the rise of ipos outside the u.s. *Journal of financial economics*, 110(3):546–573.
- Fernandez-Villaverde, J., of Economic Research., N. B., and Levintal, O. (2024). *The Distributional Effects of Asset Returns / Jesús Fernández-Villaverde, Oren Levintal*. NBER working paper series no. w32182. National Bureau of Economic Research, Cambridge, Mass.
- Fisher, J. D., Johnson, D. S., Smeeding, T. M., and Thompson, J. P. (2020). Estimating the marginal propensity to consume using the distributions of income, consumption, and wealth. *Journal of macroeconomics*, 65:103218–.

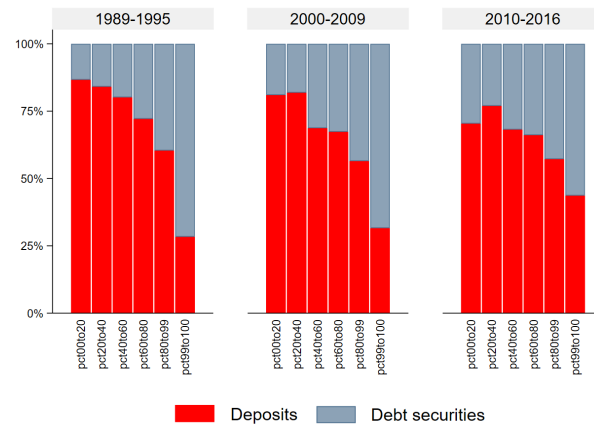
- Fontaine, J.-S. and Garcia, R. (2012). Bond liquidity premia. *The Review of financial studies*, 25(4):1207–1254.
- Gao, X., Ritter, J. R., and Zhu, Z. (2013). Where have all the ipos gone? *Journal of financial and quantitative analysis*, 48(6):1663–1692.
- Goldin, C. D. and Katz, L. F. (2008). *The race between education and technology / Claudia Goldin, Lawrence F. Katz*. Belknap Press of Harvard University Press, Cambridge, Mass.
- Gomez, M. (2025). Wealth inequality and asset prices. *The Review of economic studies*.
- Gromb, D. and Vayanos, D. (2018). The dynamics of financially constrained arbitrage. *The Journal of finance (New York)*, 73(4):1713–1750.
- Gupta, A., Sapriz, H., and Yankov, V. (2022). The collateral channel and bank credit. *IDEAS Working Paper Series from RePEc*, 22(4):1–58.
- Hall, R. E. (2016). Understanding the decline in the safe real interest rate. *NBER Working Paper Series*, pages 22196–.
- He, Z. and Krishnamurthy, A. (2013). Intermediary asset pricing. *The American economic review*, 103(2):732–770.
- Heathcote, J., Perri, F., and Violante, G. L. (2010a). Unequal we stand: An empirical analysis of economic inequality in the united states, 1967â2006. *Review of economic dynamics*, 13(1):15–51.
- Heathcote, J., Storesletten, K., and Violante, G. L. (2010b). The macroeconomic implications of rising wage inequality in the united states. *Journal of political economy*, 118(4):681–722.
- Holmstrom, B. and Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *the Quarterly Journal of economics*, 112(3):663–691.
- Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary policy according to hank. *The American economic review*, 108(3):697–743.
- Kaplan, G., Violante, G. L., and Weidner, J. (2014). The wealthy hand-to-mouth. *Brookings papers on economic activity*, 2014(1):77–138.
- Kiyotaki, N. and Moore, J. (1997). Credit cycles. *The Journal of political economy*, 105(2):211–248.

- Kopczuk, W., Saez, E., and Song, J. (2010). Earnings inequality and mobility in the united states: Evidence from social security data since 1937. *The Quarterly Journal of Economics*, 125(1):91–128.
- Kuhn, M., Schularick, M., and Steins, U. I. (2020). Income and wealth inequality in america, 1949–2016. *Journal of Political Economy*, 128(9):3469–3519.
- Meyer, B. D. and Sullivan, J. X. (2023). Consumption and income inequality in the united states since the 1960s. *Journal of Political Economy*, 131(2):247–284.
- Mian, A. R., Straub, L., and Sufi, A. (2021). What explains the decline in r^* ? rising income inequality versus demographic shifts. *Rising Income Inequality Versus Demographic Shifts (September 22, 2021)*. University of Chicago, Becker Friedman Institute for Economics Working Paper, (2021-104).
- Mian, A. R., Sufi, A., and Straub, L. (2020). The saving glut of the rich. *NBER Working Paper Series*.
- OECD (2019). Corporate bond markets in a time of unconventional monetary policy. Technical report, OECD Capital Market Series, OECD Publishing, Paris. Accessed June 2025.
- Piketty, T. and Saez, E. (2003). Income inequality in the united states, 1913–1998. *The Quarterly journal of economics*, 118(1):1–41.
- Piketty, T., Saez, E., and Zucman, G. (2018). Distributional national accounts: methods and estimates for the united states. *The Quarterly Journal of Economics*, 133(2):553–609.
- Rampini, A. A. and Viswanathan, S. (2019). Financial intermediary capital. *The Review of economic studies*, 86(1 (306)):413–455.
- Smith, M., Yagan, D., Zidar, O., and Zwick, E. (2019). Capitalists in the twenty-first century. *The Quarterly journal of economics*, 134(4):1675–1745.
- Smith, M., Zidar, O., and Zwick, E. (2023). Top wealth in america: New estimates under heterogeneous returns. *The Quarterly journal of economics*, 138(1):515–573.
- Villacorta, A. (2018). Business cycles and the balance sheets of the financial and non-financial sectors. *IDEAS Working Paper Series from RePEc*.
- Zhu, Q. (2021). Capital supply and corporate bond issuances: Evidence from mutual fund flows. *Journal of financial economics*, 141(2):551–572.

Appendix

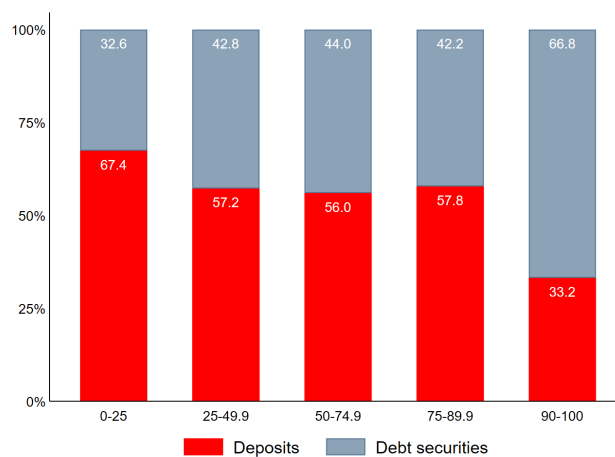
A Figures and Tables

Figure A1: Liquid wealth composition across income distribution



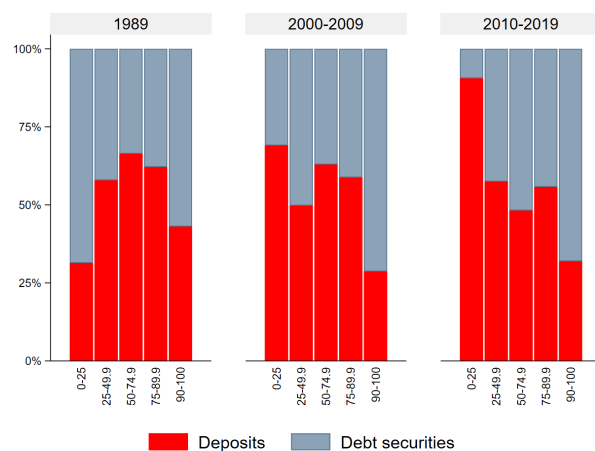
Notes: This figure shows the deposits and bond savings as a share of liquid wealth in different periods based on the Distributional Financial Accounts.

Figure A2: Average liquid wealth composition across wealth quartiles



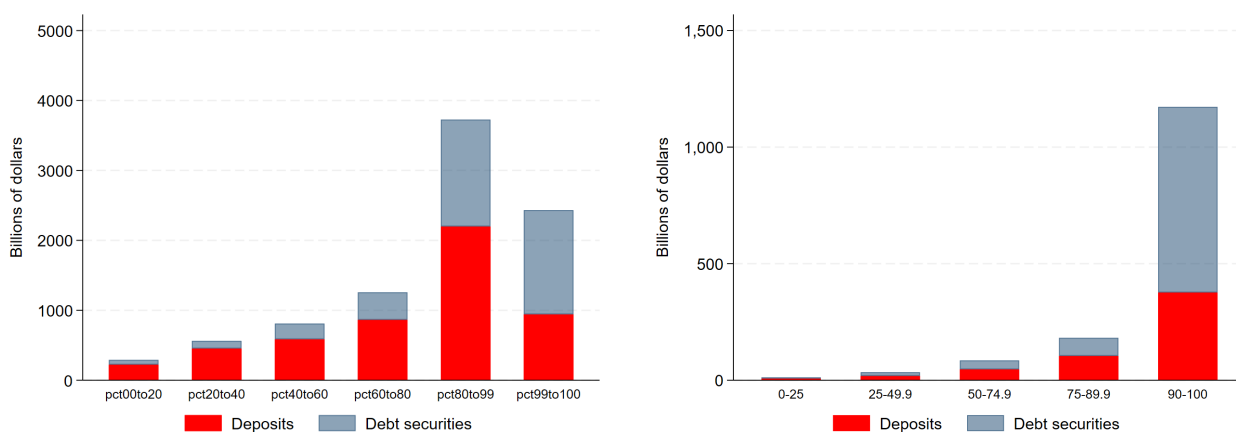
Notes: This figure shows the shares in % of deposits, such as certificates of deposit and transaction accounts, and bond savings, based on the Survey of Consumer Finances from 1989 to 2019.

Figure A3: Liquid wealth composition across wealth quartiles



Notes: This figure shows the shares in % of deposits, such as certificates of deposit and transaction accounts, and bond savings, based on the Survey of Consumer Finances.

Figure A4: Deposit and bond holdings across percentiles

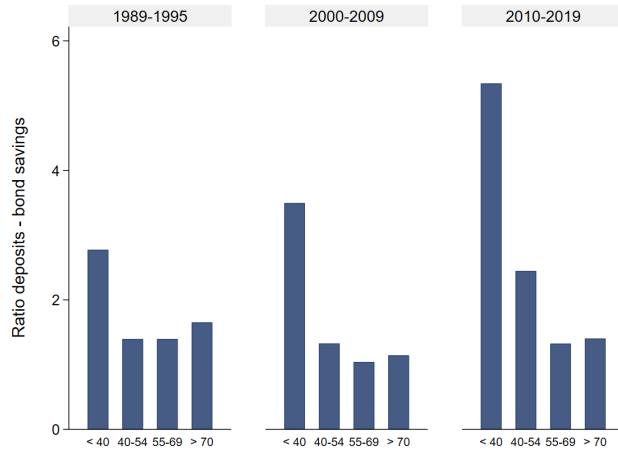


(A) By income distribution

(B) By wealth distribution

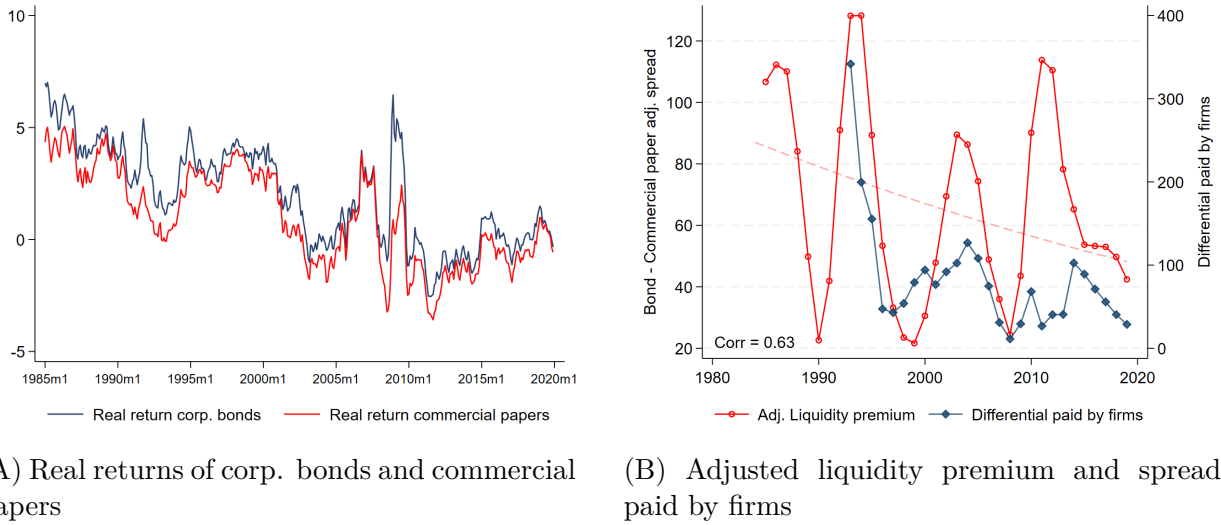
Notes: Figure A and B show the holdings of deposits and bond savings, expressed in billions of dollars, based on the Distributional Finances Account and the Survey of Consumer Finances from 1989 to 2019.

Figure A5: Deposit-to-Bonds by age-cohorts



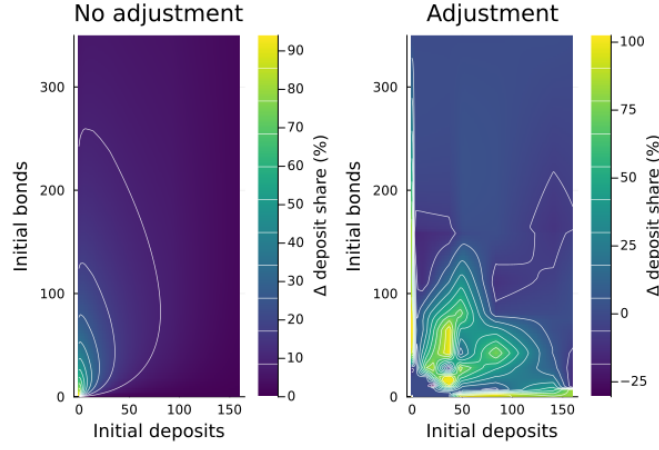
Notes: This figure shows the relative importance of deposits in different periods based on the Distributional Finance Accounts data.

Figure A6: Monthly real returns and adjusted spreads



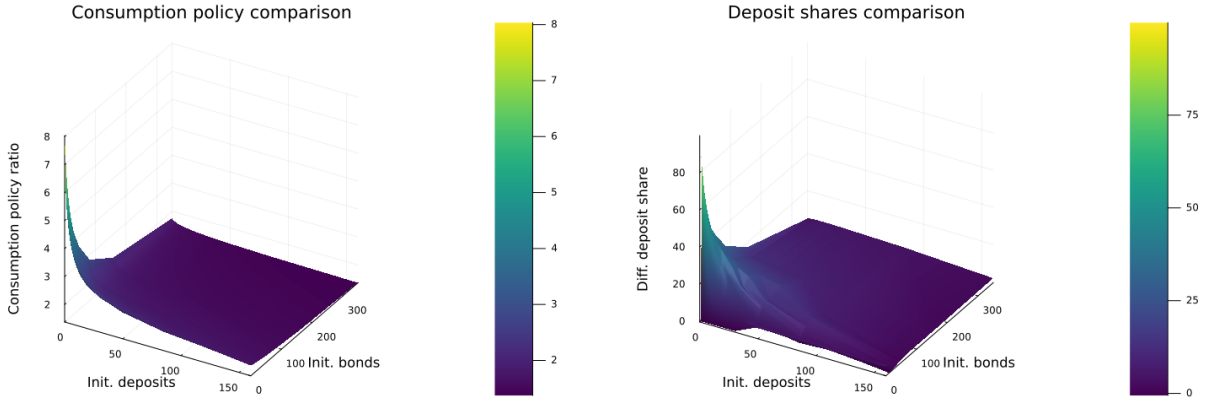
Notes: Figure A shows the returns of high-quality corporate bonds (bonds rated AAA, AA, or A with maturities of one to two years) and 3-month commercial papers offered to AA-rated firms adjusted by inflation based on monthly data published by FRED. In Figure B, the red line plots the three-year moving average of the difference between the annual average yield on high-quality corporate bonds net of the return of U.S. Treasuries at 1 year of maturity, and the interest rate on 3-month commercial papers offered to AA-rated firms net of the return of U.S. Treasuries at 3 months of maturity, based on FRED data. The blue line displays the three-year moving average of the difference between the average cost of funds from corporate bond issuance and bank loans, constructed using firm-level maturity data from Capital IQ. All series were adjusted by realized inflation.

Figure A7: Deposit shares comparison: High vs. Low productivity



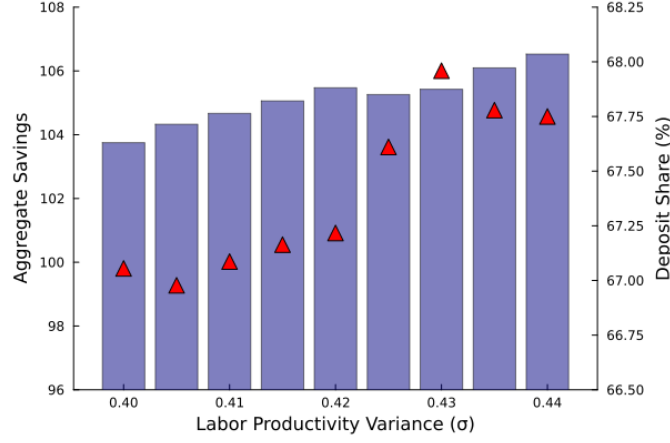
Notes: Figure A and B compare the difference in deposit shares (in percentage points) between the highest- and lowest-productivity households. The parameters used in this exercise are reported in Table 3, with a productivity ratio of $\frac{z_H}{z_L} = 22.16$. Deposit shares in both figures are computed from the interpolated policy functions for deposits and bonds, using linear splines defined over a uniform grid in both asset dimensions. Figure A illustrates the case in which households cannot adjust their bond holdings, while Figure B depicts the case where adjustment is allowed. Together, the two panels highlight how the ability to rebalance portfolios across assets modifies the sensitivity of deposit decisions to productivity differences.

Figure A8: Policy Functions: High vs. Low productivity



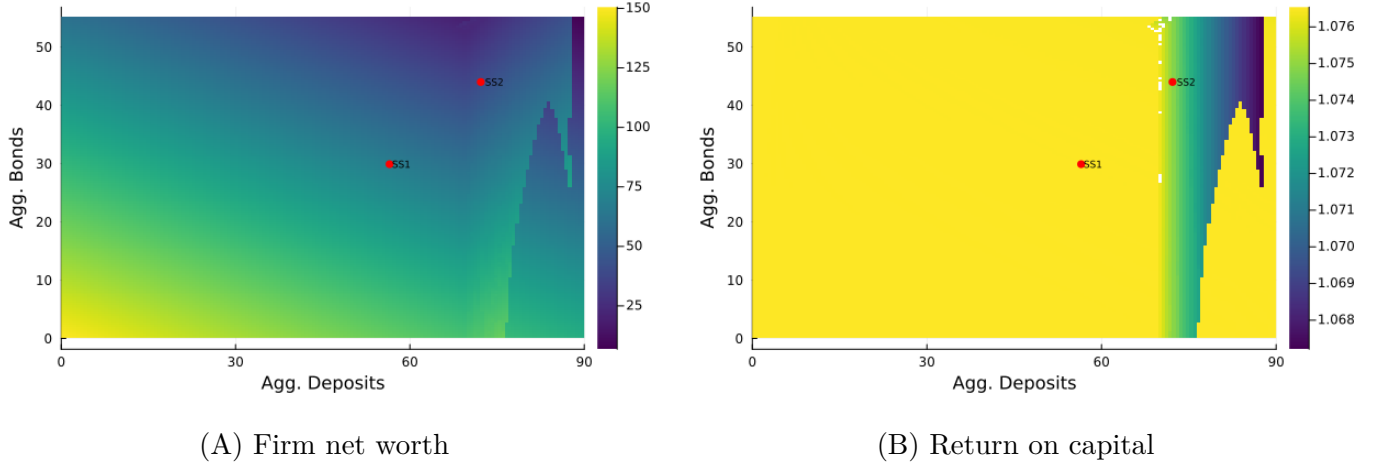
Notes: Figure A and B summarize household heterogeneity in consumption and portfolio choices. The parameters used in this exercise are reported in Table 3, with a productivity ratio of $\frac{z_H}{z_L} = 22.16$. Figure A illustrates the ratio of consumption policy of the high-productive households over low-productive households. Figure B depicts the difference in optimal deposit shares (in percentage points) between the highest- and lowest-productivity households. Optimal deposit shares are computed as the weighted average of deposits and bonds, where the weights correspond to the probability of adjustment.

Figure A9: Total savings and their composition under income inequality



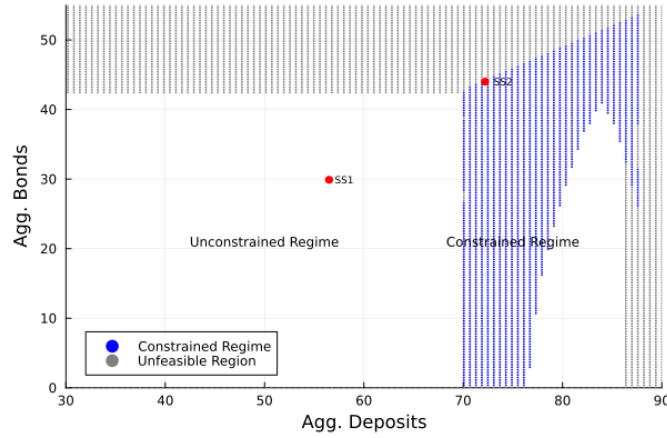
Notes: Bars (left axis) display aggregate household savings as the variance of labor productivity, σ , increases uniformly from 0.4 to 0.45. Red dots (right axis) show the share of deposits in total savings, capturing the composition of liquid wealth. The exercise uses the household parameters reported in Table 3, holding prices fixed at $r^b = 0.075$, $r^d = 0.0185$, and $W = 17.5$.

Figure A10: Net worth and return on capital



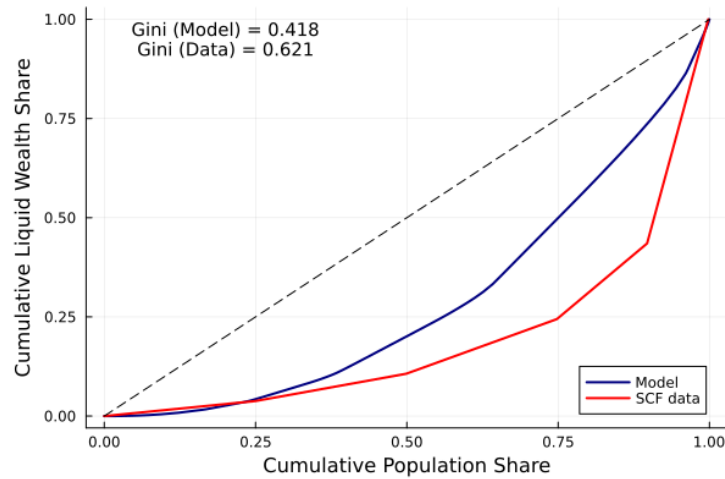
Notes: Figures (A) and (B) present heatmaps of firm net worth and the return on capital, respectively, as functions of aggregate deposits (x-axis) and aggregate bonds (y-axis). Parameters as in Table 3.

Figure A11: Unconstrained and constrained regimes



Notes: The figure distinguishes between feasible credit regimes and unfeasible allocations. White areas correspond to the unconstrained regime, while blue areas denote the constrained regime. Unfeasible allocations arise when either bond or bank collateral falls short of the required credit, in which case financial intermediaries are unwilling to lend. The exercise is based on the firm and intermediary parameters reported in Table 3.

Figure A12: Lorenz curve of liquid wealth (Model vs. Data)



Notes: The figure displays the Lorenz Curve and Gini coefficient of liquid wealth. The red and blue lines display the Lorenz Curve based on the data of deposits and bond savings of households in the 1989 SCF sample and those implied by the model, respectively.

B Households

B.1 Dynamic problem with two assets and logistic distribution of adjustment costs

$$\begin{aligned}\mathbb{E}V(d, b; z) &= V_n(d, b; z) + \int_0^{V_a - V_n} (V_a(d, b; z) - V_n(d, b; z) - \chi) f(\chi) d\chi \\ \mathbb{E}V(d, b; z) &= V_n(d, b; z) + V_a(d, b; z) F_\chi[V_a(d, b; z) - V_n(d, b; z)] \\ &\quad - V_n(d, b; z) F_\chi[V_a(d, b; z) - V_n(d, b; z)] - \int_0^{V_a - V_n} \chi f(\chi) d\chi\end{aligned}\quad (39)$$

$$\begin{aligned}\mathbb{E}V(d, b; z) &= V_n(d, b; z) + \int_0^{V_a - V_n} (V_a(d, b; z) - V_n(d, b; z) - \chi) f(\chi) d\chi \\ \mathbb{E}V(d, b; z) &= V_n(d, b; z) + V_a(d, b; z) F_\chi[V_a(d, b; z) - V_n(d, b; z)] \\ &\quad - V_n(d, b; z) F_\chi[V_a(d, b; z) - V_n(d, b; z)] - \int_0^{V_a - V_n} \chi f(\chi) d\chi\end{aligned}\quad (40)$$

Let's compact the state space $\omega := (d, b)$, then:

$$\mathbb{E}V(\omega; z) = V_a(\omega; z) \nu^*(\omega; z) + (1 - \nu^*(\omega; z)) V_n(\omega; z) - \int_0^{V_a(\omega; z) - V_n(\omega; z)} \chi f(\chi) d\chi \quad (41)$$

Then, the expected of the value function has two components: the first two terms coincides with the problem when there are fixed adjustment probabilities. The last term is the expected conditional adjustment cost.

$$\begin{aligned}\frac{\partial \mathbb{E}V(\omega; z)}{\partial \omega} &= \frac{\partial V_n(\omega; z)}{\partial \omega} + \left[\frac{\partial V_a(\omega; z)}{\partial \omega} - \frac{\partial V_n(\omega; z)}{\partial \omega} \right] [F_\chi[V_a(\omega; z) - V_n(\omega; z)]] \\ &\quad + \frac{F_\chi[V_a(\omega; z) - V_n(\omega; z)]}{\partial \omega} [V_a(\omega; z) - V_n(\omega; z)] - \frac{\partial}{\partial \omega} \int_0^{V_a(\omega; z) - V_n(\omega; z)} \chi f(\chi) d\chi\end{aligned}\quad (42)$$

Applying Leibniz's rule and rearranging terms, I obtain:

$$\begin{aligned}\frac{\partial \mathbb{E}V(\omega; z)}{\partial \omega} &= \frac{\partial V_a(\omega; z)}{\partial \omega} [F_\chi[V_a(\omega; z) - V_n(\omega; z)]] \\ &\quad + \frac{\partial V_n(\omega; z)}{\partial \omega} [1 - F_\chi[V_a(\omega; z) - V_n(\omega; z)]] \frac{\partial \mathbb{E}V(\omega; z)}{\partial \omega} \\ &= \nu^*(\omega; z) \frac{\partial V_a(\omega; z)}{\partial \omega} + (1 - \nu^*(\omega; z)) \frac{\partial V_n(\omega; z)}{\partial \omega}\end{aligned}\quad (43)$$

In simple words, F.O.C of the expected value function with fixed adjustment probabilities and a model with state-dependent adjustment probabilities are the same.

Therefore, the Euler Equations (for interior solutions) around the optimal policies: c_i^* , d_i^* and b^* are the optimal policies of consumption, deposits and bond holdings where $i \in \{a, n\}$, respectively.

$$\begin{aligned}
[b^*] \frac{\partial u(c_a^*)}{\partial c} &= \beta \mathbb{E} \left[\nu \frac{\partial V_a(d_a^*, b^*, z')}{\partial b} + (1 - \nu) \frac{\partial V_n(d_a^*, b^*, z')}{\partial b} \right] \\
[d_a^*] \frac{\partial u(c_a^*)}{\partial c} &= \beta \mathbb{E} \left[\nu \frac{\partial V_a(d_a^*, b^*, z')}{\partial d} + (1 - \nu) \frac{\partial V_n(d_a^*, b^*, z')}{\partial d} \right] \\
[d_n^*] \frac{\partial u(c_n^*)}{\partial c} &= \beta \mathbb{E} \left[\nu \frac{\partial V_a(d_n^*, b, z')}{\partial d} + (1 - \nu) \frac{\partial V_n(d_n^*, b, z')}{\partial d} \right]
\end{aligned} \tag{44}$$

Using the Envelope Theorem conditions (EC), I get:

$$\begin{aligned}
[b] \frac{\partial V_a(d, b, z)}{\partial b} &= \frac{\partial u(c_a^*(d, b, z))}{\partial c} (1 + r^b) \\
[d] \frac{\partial V_a(d, b, z)}{\partial d} &= \frac{\partial u(c_a^*(d, b, z))}{\partial c} (1 + r^d) \\
[b] \frac{\partial V_n(d, b, z)}{\partial b} &= \frac{\partial u(c_n^*(d, b, z))}{\partial c} (1 + r^b) + \beta v \mathbb{E} \left[\frac{\partial V_a(d_n^*(d, b, z), b, z')}{\partial b} \right] \\
&\quad + \beta (1 - \nu) \mathbb{E} \left[\frac{\partial V_n(d_n^*(d, b, z), b, z')}{\partial b} \right] \\
&= \frac{\partial u(c_n^*(d, b, z))}{\partial c} (1 + r^b) + \beta v (1 + r^b) \mathbb{E} \left[\frac{\partial u(c_a^*(d_n^*(d, b, z), b, z'))}{\partial c} \right] \\
&\quad + \beta (1 - \nu) \mathbb{E} \left[\frac{\partial V_n(d_n^*(d, b, z), b, z')}{\partial b} \right] \\
[d] \frac{\partial V_n(d, b, z)}{\partial d} &= \frac{\partial u(c_n^*(d, b, z))}{\partial c} (1 + r^d)
\end{aligned} \tag{45}$$

Plugging Equation 45 into Equation 44:

$$\begin{aligned}
[b^*] \frac{\partial u(c_a^*)}{\partial c} &= \beta \mathbb{E} \left[\nu \frac{\partial u(c_a^*(d_a^*, b^*, z'))}{\partial c} (1 + r^b) + (1 - \nu) \frac{\partial V_n(d_a^*, b^*, z')}{\partial b} \right] \\
[d_a^*] \frac{\partial u(c_a^*)}{\partial c} &= \beta \mathbb{E} \left((1 + r^d) \left[\nu \frac{\partial u(c_a^*(d_a^*, b^*, z'))}{\partial c} + (1 - \nu) \frac{\partial u(c_n^*(d_a^*, b^*, z'))}{\partial c} \right] \right) \\
[d_n^*] \frac{\partial u(c_n^*)}{\partial c} &= \beta \mathbb{E} \left((1 + r^d) \left[\nu \frac{\partial u(c_a^*(d_n^*, b, z'))}{\partial c} + (1 - \nu) \frac{\partial u(c_n^*(d_n^*, b, z'))}{\partial c} \right] \right)
\end{aligned} \tag{46}$$

Finally, to compute the expected adjustment costs $AC(v; \mu_x, \sigma_x)$, assuming that F_χ is a logistic distribution defined by:

$$\begin{aligned}
F(\chi) &= \frac{1}{1 + \exp\left\{-\frac{\chi - \mu_x}{\sigma_x}\right\}} \\
1 + \exp\left\{-\frac{\chi - \mu_x}{\sigma_x}\right\} &= \frac{1}{F(\chi)} \Rightarrow -\frac{\chi - \mu_x}{\sigma_x} = \ln\left(\frac{1}{F(\chi)} - 1\right) \\
\Rightarrow \frac{\mu_x}{\sigma_x} &= \ln\left(\frac{1}{p} - 1\right) + \frac{\chi}{\sigma_x} \Rightarrow \chi = \sigma_x \left[\frac{\mu_x}{\sigma_x} - \ln\left(\frac{1-p}{p}\right) \right].
\end{aligned} \tag{47}$$

Thus, I get that $\chi = \mu_x + \sigma_x [\ln(p) - \ln(1-p)]$. Hence, the expected conditional adjustment cost is:

$$\begin{aligned}
\underbrace{\int_0^{V_a(\omega; z) - V_n(\omega; z)} \chi f(\chi) d\chi}_{\equiv AC(v; \mu_x, \sigma_x)} &= \int_0^{F^{-1}\left(\underbrace{V_a(\omega; z) - V_n(\omega; z)}_{=v}\right)} \chi \underbrace{dF(\chi)}_{=dp} \\
&= \int_0^v F^{-1}(p) \underbrace{dp}_{=dp} = \int_0^v \mu_x + \sigma_x [\ln(p) - \ln(1-p)] dp
\end{aligned} \tag{48}$$

Recall that $\int \ln x dx = x(\ln x - 1) + c$, then

$$\int_0^{V_a(\omega; z) - V_n(\omega; z)} \chi f(\chi) d\chi = \mu_x v + \left(\sigma_x \left[p(\ln p - 1) - \int_0^v \ln(1-p) dp \right] \right) \Big|_0^v \tag{49}$$

Solve $\int_0^v \ln(1-p) dp$ using integration by parts where $u = \ln(1-p)$ and $dv = dp$. Then $du = -\frac{1}{1-p} dp$ and $v = p$, therefore $\int_0^v \ln(1-p) dp = p \ln(1-p) + \int_0^v p \frac{1}{1-p} dp$.

Therefore, for every $v \in [0, 1)$

$$\int_0^{V_a(\omega; z) - V_n(\omega; z)} \chi f(\chi) d\chi = \mu_x v + \sigma_x (v \ln v + (1-v) \ln(1-v)) \tag{50}$$

C Firms

C.1 Labor market

The labor market is competitive, and firms take the wage w_t as given. Wages fall to zero if the firm fails, since contracts are contingent on firm productivity. Thus, for a given level of capital k_{jt+1} , firms maximize cash flows by choosing labor to solve:

$$y_{jt} = \max_{l_{jt}} \mathbb{E}_t [\omega_{jt}] \left(A k_{jt+1}^\alpha l_{jt}^{1-\alpha} - w_t l_{jt} \right) \quad (51)$$

where the optimal labor decision is linear in capital since $l_{jt}^*(k_{jt+1}) = \left(\frac{A(1-\alpha)}{w_t} \right)^{\frac{1}{\alpha}} k_{jt+1}$ and each firm's gross profits are defined by:

$$\pi(k_{jt+1}, \omega_{jt}) = \omega_{jt} \alpha A^{\frac{1}{\alpha}} \left(\frac{1-\alpha}{w_t} \right)^{\frac{1-\alpha}{\alpha}} k_{jt+1} \quad (52)$$

Therefore, the marginal return of a firm's capital will be influenced by idiosyncratic productivity ω_{jt} and also depends on wages and the common technology factor. As a result, the marginal product of capital is identical across all firms, given by $\frac{\partial \pi_{jt}}{\partial k_{jt+1}} = \omega_{jt} \alpha A^{\frac{1}{\alpha}} \left(\frac{1-\alpha}{w_t} \right)^{\frac{1-\alpha}{\alpha}}$. Moreover, in the labor market, the aggregate demand for labor is:

$$\int_{j \in \mathbb{J}} l_{jt}^*(k_{jt+1}) dj = \left(\frac{A(1-\alpha)}{w_t} \right)^{\frac{1}{\alpha}} K_{t+1} \quad (53)$$

and the wage w_t clears the market of labor. Also, given the aggregate fixed labor supply from households, the equilibrium wage solves:

$$\left(\frac{A(1-\alpha)}{w_t} \right)^{\frac{1}{\alpha}} K_{t+1} = 1 \quad (54)$$

then wages depend on aggregate capital, and by substituting Equation 54 into Equation 52, I obtain:

$$\frac{\partial \pi_{jt}}{\partial k_{jt+1}} = \begin{cases} \alpha A K_{t+1}^{\alpha-1} & \text{with } p_H \\ 0 & \text{with } 1 - p_H \end{cases} \quad (55)$$

This reflects the fact that, from an individual firm's perspective, profits are linear in capital, whereas at the aggregate level they decline with the total capital stock.

C.2 Firm's optimal borrowing decision

Solving the dynamic problem of firms:

$$\begin{aligned} V_t^F(n_{jt}) = & \tau \Lambda \mathbb{E}_t [R_{jt}^K k_{jt+1} - R_t^B L_{jt}^B - R_{jt}^D L_{jt}^D] \\ & + (1 - \tau) \Lambda \mathbb{E}_t [V_{t+1}^F(n_{jt+1})] + \lambda_{bt} \left((1 - \kappa) k_{jt+1} - R_t^B L_{jt}^B \right) \\ & + \lambda_{dt} \left(\phi \kappa k_{jt+1} - R_{jt}^D L_{jt}^D \right) \end{aligned}$$

Conjecture: $V_t^F(n_{jt}) = \theta_t n_{jt}$ and policies $L_{jt}^B = \tilde{b} n_{jt}$, $L_{jt}^D = \tilde{d} n_{jt}$ then $k_{jt+1} = (1 + \tilde{b} + \tilde{d}) n_{jt}$. Therefore,

$$\begin{aligned} V_t^F(n_{jt}) = & \theta_t n_{jt} = \tau \Lambda \mathbb{E}_t [R_{jt}^K (1 + \tilde{b} + \tilde{d}) - R_t^B \tilde{b} - R_{jt}^D \tilde{d}] n_{jt} \\ & + (1 - \tau) \Lambda \mathbb{E}_t [\theta_{t+1} (R_{jt}^K (1 + \tilde{b} + \tilde{d}) - R_t^B \tilde{b} - R_{jt}^D \tilde{d})] n_{jt} \\ & + \lambda_{bt} \left((1 - \kappa) (1 + \tilde{b} + \tilde{d}) - R_t^B \tilde{b} \right) n_{jt} + \lambda_{dt} \left(\phi \kappa (1 + \tilde{b} + \tilde{d}) - R_{jt}^D \tilde{d} \right) n_{jt} \end{aligned}$$

where $n_{jt+1} = R_{jt}^K (1 + \tilde{b} + \tilde{d}) n_{jt} - R_t^B \tilde{b} n_{jt} - R_{jt}^D \tilde{d} n_{jt}$. Moreover, R_{jt}^K and R_{jt}^D are drawn from a common distribution, independent across firms, then $\mathbb{E}_t R_{jt}^K = \mathbb{E}_t R_t^K$ and $\mathbb{E}_t R_{jt}^D = \mathbb{E}_t R_t^D$. Suppose θ_{T+1} is known and independent of firm j (e.g., terminal value or steady state). Then, the expectation $\mathbb{E}_T[\theta_{T+1} \cdot X_{jT}]$ can be evaluated as:

$$\mathbb{E}_T[\theta_{T+1} \cdot X_{jT}] = \theta_{T+1} \cdot \mathbb{E}_T[X_{jT}] = \theta_{T+1} \cdot \mathbb{E}_T[X_T]$$

since θ_{t+1} is not firm-specific, and the shocks to X_{jt} are identically distributed across firms. Thus, θ_t computed from this recursion is also not firm-specific.

$$\begin{aligned} \theta_t = & \tau \Lambda \mathbb{E}_t [R_t^K (1 + \tilde{b} + \tilde{d}) - R_t^B \tilde{b} - R_t^D \tilde{d}] \\ & + (1 - \tau) \Lambda \mathbb{E}_t [\theta_{t+1} (R_t^K (1 + \tilde{b} + \tilde{d}) - R_t^B \tilde{b} - R_t^D \tilde{d})] \\ & + \lambda_{bt} \left((1 - \kappa) (1 + \tilde{b} + \tilde{d}) - R_t^B \tilde{b} \right) + \lambda_{dt} \left(\phi \kappa (1 + \tilde{b} + \tilde{d}) - R_t^D \tilde{d} \right) \end{aligned} \tag{56}$$

which proves the conjecture. Then, F.O.Cs are defined by:

$$\begin{aligned}
[\tilde{b}] \quad & \tau\Lambda(\mathbb{E}_t R_{jt}^K - R_t^B) + (1 - \tau)\Lambda\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)] \\
& + \lambda_{bt} ((1 - \delta)(1 - \kappa) - R_t^B) + \lambda_{dt} (\phi\kappa(1 - \delta)) = 0
\end{aligned} \tag{57}$$

$$\begin{aligned}
[\tilde{d}] \quad & \tau\Lambda\mathbb{E}_t (R_{jt}^K - R_{jt}^D) + (1 - \tau)\Lambda\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_{jt}^D)] \\
& + \lambda_{bt} ((1 - \delta)(1 - \kappa)) + \lambda_{dt} (\phi\kappa(1 - \delta) - R_{jt}^D) = 0
\end{aligned} \tag{58}$$

both conditions are equal to zero in equilibrium, otherwise if they are greater than zero, firms will borrow infinitely amount of resources, and if it is lower than zero, firms will borrow nothing.

Case 1: Both financial constraints are not binding. Under this scenario $\lambda_{bt}, \lambda_{dt} = 0$

$$[\tilde{b}] \quad \tau\Lambda(\mathbb{E}_t R_{jt}^K - R_t^B) + (1 - \tau)\Lambda\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)] = 0$$

$$[\tilde{d}] \quad \tau\Lambda\mathbb{E}_t (R_{jt}^K - R_{jt}^D) + (1 - \tau)\Lambda\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_{jt}^D)] = 0$$

To satisfy both conditions, then $\mathbb{E}_t R_{jt}^K = R_t^B$, $\mathbb{E}_t (R_{jt}^K - R_{jt}^D) = 0$, $\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_{jt}^D)] = cov(\theta_{t+1}, R_{jt}^K - R_{jt}^D) + \mathbb{E}_t [\theta_{t+1}] \mathbb{E}_t [R_{jt}^K - R_{jt}^D] = 0$ and $\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)] = cov(\theta_{t+1}, R_{jt}^K - R_t^B) + \mathbb{E}_t [\theta_{t+1}] \mathbb{E}_t [R_{jt}^K - R_t^B] = 0$

Case 2: Only the bond financial constraint is binding. Under this scenario $\lambda_{dt} = 0$ and $\lambda_{bt} > 0$

$$[\tilde{b}] \quad \lambda_{bt} = \frac{\tau\Lambda(\mathbb{E}_t R_{jt}^K - R_t^B) + (1 - \tau)\Lambda\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)]}{R_t^B - (1 - \delta)(1 - \kappa)}$$

$$\begin{aligned}
[\tilde{d}] \quad & \tau\Lambda\mathbb{E}_t (R_{jt}^K - R_{jt}^D) + (1 - \tau)\Lambda\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_{jt}^D)] \\
& + \lambda_{bt} ((1 - \delta)(1 - \kappa)) = 0
\end{aligned}$$

Combining these two conditions,

$$\begin{aligned}
& \tau\mathbb{E}_t (R_{jt}^K - R_{jt}^D) + (1 - \tau)\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_{jt}^D)] \\
& + \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \left(\tau(\mathbb{E}_t R_{jt}^K - R_t^B) + (1 - \tau)\mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)] \right) = 0
\end{aligned}$$

$$\begin{aligned} & \tau \mathbb{E}_t(R_{jt}^K - R_{jt}^D) + (1 - \tau) \left(\text{cov}(\theta_{t+1}, R_{jt}^K - R_{jt}^D) + \mathbb{E}_t(R_{jt}^K - R_{jt}^D) \mathbb{E}_t \theta_{t+1} \right) \\ & + \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \left(\tau (\mathbb{E}_t R_{jt}^K - R_t^B) + (1 - \tau) \left(\text{cov}(\theta_{t+1}, R_{jt}^K - R_t^B) + \mathbb{E}_t(R_{jt}^K - R_t^B) \mathbb{E}_t \theta_{t+1} \right) \right) = 0 \end{aligned}$$

$$\begin{aligned} & \mathbb{E}_t \left(\frac{R_{jt}^K}{1 - \frac{(1 - \delta)(1 - \kappa)}{R_t^B}} - R_{jt}^D \right) (\tau + (1 - \tau) \mathbb{E}_t \theta_{t+1}) \\ & - \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \left(R_t^B (\tau + (1 + \tau) \mathbb{E}_t \theta_{t+1}) \right) \\ & + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_{jt}^K - R_{jt}^D) + \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \text{cov}(\theta_{t+1}, R_t^K - R_t^B) \right] = 0 \end{aligned}$$

$$\begin{aligned} & \mathbb{E}_t \left(\frac{R_{jt}^K}{1 - \frac{(1 - \delta)(1 - \kappa)}{R_t^B}} - R_{jt}^D \right) (\tau + (1 - \tau) \mathbb{E}_t \theta_{t+1}) \\ & - \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \left(R_t^B (\tau + (1 + \tau) \mathbb{E}_t \theta_{t+1}) \right) \\ & + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_t^K) - \text{cov}(\theta_{t+1}, R_t^D) + \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} (\text{cov}(\theta_{t+1}, R_t^K) - \text{cov}(\theta_{t+1}, R_t^B)) \right] = 0 \end{aligned}$$

$$\begin{aligned} & \mathbb{E}_t \left(\frac{R_{jt}^K}{1 - \frac{(1 - \delta)(1 - \kappa)}{R_t^B}} - R_{jt}^D \right) (\tau + (1 - \tau) \mathbb{E}_t \theta_{t+1}) \\ & - \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \left(R_t^B (\tau + (1 + \tau) \mathbb{E}_t \theta_{t+1}) \right) \\ & + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_t^K) \left[1 + \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \right] - \text{cov}(\theta_{t+1}, R_t^D) \right] = 0 \end{aligned}$$

$$\begin{aligned} & \mathbb{E}_t \left(\frac{R_{jt}^K - (1 - \delta)(1 - \kappa)}{1 - \frac{(1 - \delta)(1 - \kappa)}{R_t^B}} - R_{jt}^D \right) (\tau + (1 - \tau) \mathbb{E}_t \theta_{t+1}) \\ & + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_{jt}^K) \left[1 + \frac{(1 - \delta)(1 - \kappa)}{R_t^B - (1 - \delta)(1 - \kappa)} \right] - \text{cov}(\theta_{t+1}, R_t^D) \right] = 0 \end{aligned}$$

Recall that $R_t^K = \omega_{jt}(\alpha A_t K_{t+1}^{\alpha-1} + (1 - \delta)\kappa) + (1 - \delta)(1 - \kappa)$

$$\begin{aligned}
& \mathbb{E}_t \left(\omega_{jt} \left(\frac{\alpha A_t K_{t+1}^{\alpha-1} + (1-\delta)\kappa}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} - R_{jt}^D \right) \right) (\tau + (1-\tau)\mathbb{E}_t \theta_{t+1}) \\
& + (1-\tau) \left[\text{cov}(\theta_{t+1}, R_t^K) \left[\frac{1}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} \right] - \text{cov}(\theta_{t+1}, R_t^D) \right] = 0 \\
& \mathbb{E}_t \left(\omega_t \left(\frac{\alpha A_t K_{t+1}^{\alpha-1} + (1-\delta)\kappa}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} - R_t^D \right) \right) (\tau + (1-\tau)\mathbb{E}_t \theta_{t+1}) \\
& + (1-\tau) \left[\text{cov}(\theta_{t+1}, \omega_{jt}) \left(\frac{\alpha A_t K_{t+1}^{\alpha-1} + (1-\delta)\kappa}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} - R_t^D \right) \right] = 0
\end{aligned} \tag{59}$$

In the model, the covariance term $\text{cov}(\theta_{t+1}, \omega_{jt})$ vanishes since the individual idiosyncratic shocks ω_{jt} are independent of aggregate prices. Since θ_{t+1} depends only on R_t^B, R_t^D , and R_t^K which are determined by aggregate states. With a continuum of firms, an individual draw has measure zero impact on equilibrium prices, ensuring conditional independence. Thus, the covariance term drops out, and the relevant expectation reduces to the first component in Equation 59.

Case 3: Only the risky bank loan financial constraint is binding. Under this scenario $\lambda_{bt} = 0$ and $\lambda_{dt} > 0$

$$[\tilde{b}] \quad \tau \Lambda \mathbb{E}_t (R_{jt}^K - R_t^B) + (1-\tau) \Lambda \mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)] + \lambda_{dt} (\phi \kappa (1-\delta)) = 0$$

$$[\tilde{d}] \quad \lambda_{dt} = \frac{\tau \Lambda \mathbb{E}_t (R_{jt}^K - \omega_{jt} R_t^D) + (1-\tau) \Lambda \mathbb{E}_t [\theta_{t+1} (R_{jt}^K - \omega_{jt} R_t^D)]}{\omega_{jt} R_t^D - \phi \kappa (1-\delta)}$$

Combining these two conditions,

$$\begin{aligned}
& \tau \Lambda \mathbb{E}_t (R_{jt}^K - R_t^B) + (1-\tau) \Lambda \mathbb{E}_t [\theta_{t+1} (R_{jt}^K - R_t^B)] \\
& + \frac{\phi \kappa (1-\delta)}{\omega_{jt} R_t^D - \phi \kappa (1-\delta)} \left(\tau \Lambda \mathbb{E}_t (R_{jt}^K - \omega_{jt} R_t^D) + (1-\tau) \Lambda \mathbb{E}_t [\theta_{t+1} (R_{jt}^K - \omega_{jt} R_t^D)] \right) = 0
\end{aligned}$$

$$\begin{aligned}
& \tau \mathbb{E}_t(R_{jt}^K \left[\frac{\omega_{jt} R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] - R_t^B) + (1 - \tau) \mathbb{E}_t(\theta_{t+1}) \mathbb{E}_t(R_{jt}^K \left[\frac{\omega_{jt} R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] - R_t^B) \\
& \quad + (1 - \tau) \text{cov}(\theta_{t+1}, R_{jt}^K - R_t^B) \\
& + \frac{\phi \kappa (1 - \delta)}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \left(-\tau \mathbb{E}_t(\omega_{jt} R_t^D) + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_{jt}^K - \omega_{jt} R_t^D) - \mathbb{E}_t(\theta_{t+1}) \mathbb{E}_t(\omega_{jt} R_t^D) \right] \right) = 0
\end{aligned}$$

$$\begin{aligned}
& \mathbb{E}_t \left(R_{jt}^K \left[\frac{\omega_{jt} R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] - R_t^B \right) (\tau + (1 - \tau) \mathbb{E}_t(\theta_{t+1})) \\
& \quad - \frac{\phi \kappa (1 - \delta)}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \mathbb{E}_t(\omega_{jt} R_t^D) (\tau + (1 - \tau) \mathbb{E}_t(\theta_{t+1})) \\
& + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_{jt}^K - R_t^B) + \frac{\phi \kappa (1 - \delta)}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \text{cov}(\theta_{t+1}, R_{jt}^K - \omega_{jt} R_t^D) \right] = 0
\end{aligned}$$

$$\begin{aligned}
& \left[\mathbb{E}_t \left((R_{jt}^K - \phi \kappa (1 - \delta)) \left[\frac{\omega_{jt} R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] - R_t^B \right) \right] (\tau + (1 - \tau) \mathbb{E}_t(\theta_{t+1})) \\
& + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_{jt}^K) \left[\frac{\omega_{jt} R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] - \frac{\phi \kappa (1 - \delta)}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \text{cov}(\theta_{t+1}, \omega_{jt} R_t^D) \right] = 0
\end{aligned}$$

$$\begin{aligned}
& \mathbb{E}_t \left(\frac{\omega_{jt} (R_{jt}^K - \phi \kappa (1 - \delta))}{\omega_{jt} - \frac{\phi \kappa (1 - \delta)}{R_t^D}} - R_t^B \right) (\tau + (1 - \tau) \mathbb{E}_t(\theta_{t+1})) \\
& + (1 - \tau) \left[\text{cov}(\theta_{t+1}, R_{jt}^K) \left[\frac{\omega_{jt} R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] - \frac{\phi \kappa (1 - \delta)}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \text{cov}(\theta_{t+1}, \omega_{jt} R_t^D) \right] = 0
\end{aligned}$$

Recall that $R_{jt}^K = \omega_{jt}(\alpha A_t K_{t+1}^{\alpha-1} + (1 - \delta)\kappa) + (1 - \delta)(1 - \kappa)$

$$\begin{aligned}
& \mathbb{E}_t \left(\frac{\omega_{jt} (R_{jt}^K - \phi \kappa (1 - \delta))}{\omega_{jt} - \frac{\phi \kappa (1 - \delta)}{R_t^D}} - R_t^B \right) (\tau + (1 - \tau) \mathbb{E}_t(\theta_{t+1})) \\
& + (1 - \tau) \text{cov}(\theta_{t+1}, \omega_{jt}) \left[\frac{\omega_{jt} R_t^D (\alpha A_t K_{t+1}^{\alpha-1} + (1 - \delta)\kappa) - \phi \kappa (1 - \delta) R_t^D}{\omega_{jt} R_t^D - \phi \kappa (1 - \delta)} \right] = 0
\end{aligned}$$

$$\begin{aligned} & \mathbb{E}_t \left(\frac{R_{jt}^K - \phi\kappa(1-\delta)}{1 - \frac{\phi\kappa(1-\delta)}{\omega_{jt}R_t^D}} - R_t^B \right) (\tau + (1-\tau)\mathbb{E}_t(\theta_{t+1})) \\ & + (1-\tau)cov(\theta_{t+1}, \omega_{jt}) \left[\frac{\alpha A_t K_{t+1}^{\alpha-1} + (1-\delta)\kappa - \phi\kappa(1-\delta)}{1 - \frac{\phi\kappa(1-\delta)}{\omega_{jt}R_t^D}} \right] = 0 \end{aligned} \quad (60)$$

Similarly to Case 1, the covariance term $cov(\theta_{t+1}, \omega_{jt})$ vanishes since the individual idiosyncratic shocks ω_{jt} are independent of aggregate prices. Thus, the covariance term drops out, and the relevant expectation reduces to the first component in Equation 60.

Case 4: Both financial constraints are binding. In this scenario $\lambda_{bt}, \lambda_{dt} > 0$. This implies that the conditions 59 and 60 are strictly positive, i.e.,

$$\frac{\alpha A_t K_{t+1}^{\alpha-1} + (1-\delta)\kappa}{1 - \frac{(1-\delta)(1-\kappa)}{R_t^B}} - R_t^D > 0; \quad \frac{R_{jt}^K - \phi\kappa(1-\delta)}{1 - \frac{\phi\kappa(1-\delta)}{\omega_{jt}R_t^D}} - R_t^B > 0 \quad (61)$$

These inequalities indicate that the relative return to capital, measured against each alternative debt instrument, exceeds the respective borrowing rate. In other words, both bank loans and bonds offer financing at rates below the effective marginal value of capital. As a result, firms optimally choose to hold and utilize both forms of debt simultaneously.

D Banks

Introducing unsecured consumer credit: Households can access unsecured consumer credit ($d_{it} < 0$), subject to an intermediation cost $\hat{r}$ per unit of credit and a monitoring cost ϵ incurred by banks for each unit of household debt. The monitoring technology is intended to mitigate default risk. Under this setup, the profits of banks from consumer lending are given by:

$$\pi_t^D = \int_j R_{jt}^D L_{jt}^D dj - (1 + r_t^d) \int_i \mathbb{I}_{d_{it} \geq 0} |d_{it}| d_i + (1 + r_t^d + \hat{r} - \epsilon) \int_i \mathbb{I}_{d_{it} < 0} |d_{it}| d_i - \hat{c} L_t^D \geq 0 \quad (62)$$

I define the composition of total deposits based on the sign of individual household positions. Specifically, the share of total deposits that are non-negative and negative are denoted by $S_t^+ = \frac{\int_i \mathbb{I}_{d_{it} \geq 0} |d_{it}| d_i}{\int_i |d_{it}| d_i}$ and $S_t^- = \frac{\int_i \mathbb{I}_{d_{it} < 0} |d_{it}| d_i}{\int_i |d_{it}| d_i}$, respectively. By construction, these shares satisfy $S_t^+ + S_t^- = 1$.

In a competitive market, where bank profits from deposit intermediation satisfy $\pi_t^D = 0$, the following condition must hold:

$$\pi_t^D = p_H R_t^D L_t^D - \left((1 + r_t^d) S_t^+ - (1 + r^d + \hat{r} - \epsilon) S_t^- \right) \int_i |d_{it}| d_i - \hat{c} L_t^D \geq 0$$

where $L_t^D = D_t$ and $D_t = S_t^+ \int_i |d_{it}| d_i - S_t^- \int_i |d_{it}| d_i$

$$(p_H R_t^D - \hat{c}) (S_t^+ - S_t^-) = (1 + r_t^d) S_t^+ - (1 + r^d + \hat{r} - \epsilon) S_t^- \quad (63)$$

For simplicity, I assume that the extra interest rate charged on household borrowing $\hat{r}$, fully covers the monitoring cost ϵ associated with unsecured consumer credit. Moreover, given that monitoring firms is also costly, the supply of risky bank loans in the presence of unsecured consumer credit is defined by:

$$p_H R_t^D = 1 + r_t^d + \hat{c} \quad (64)$$

Equation 64 indicates that, all else equal, a rise in $\hat{c}$ requires a lower deposit rate r_t^d to restore balance.

D.1 Moral hazard and financial contracts

This section describes the financial contracts between firms, banks, and bondholders, highlighting how monitoring and incentive constraints shape short-term financing arrangements.

The binomial nature of the firm's asset value (collateral) allows monitoring to play a central role in capital intermediation: it prevents the firm from diverting more than a fraction ϕ of its risky capital. The parameter ϕ measures the effectiveness of monitoring in curbing misreporting. A higher ϕ reflects stronger enforcement, making it more difficult for firms to divert resources.

By including banks in financial contracts, firms gain access to greater external financing, but at the cost of tighter discipline. Monitoring restricts the firm's ability to misreport or expropriate funds, and to ensure truthful reporting, banks impose the following incentive compatibility constraint:

$$\begin{aligned}
& \pi(k_{jt+1}, \omega_{jt} = 1) + (1 - \delta)k_{jt+1} - R_t^B L_{jt}^B - R_t^D (\omega_{jt} = 1) L_{jt}^D \geq \\
& \pi(k_{jt+1}, \omega_{jt} = 0 | \omega_{jt} = 1) + (1 - \delta)k_{jt+1} - R_t^B L_{jt}^B - R_t^D (\omega_{jt} = 0 | \omega_{jt} = 1) L_{jt}^D - \phi(1 - \delta)\kappa k_{jt+1} \\
& \Rightarrow R_t^D (\omega_{jt} = 1) L_{jt}^D - R_t^D (\omega_{jt} = 0 | \omega_{jt} = 1) L_{jt}^D \leq \phi\kappa(1 - \delta)k_{jt+1}
\end{aligned} \tag{65}$$

Thus, bank monitoring imposes a constraint on firms, as they lose a fraction $\phi \in (0, 1)$ of the risky capital due to monitoring enforcement. Banks accept risky debt, but only up to the fraction of risky capital that can be effectively monitored. Additionally, since bondholders (households) can't monitor firms, they only accept risk-free contracts, requiring full collateralization. Bondholders also recognize that banks have a senior claim in failure states. To ensure bondholders' risk-free repayment, they impose the following feasibility constraint:

$$(1 - \delta)(1 - \kappa)k_{jt} \geq R_t^B L_{jt}^B + R_t^D (\omega_{jt} = 0) L_{jt}^D \tag{66}$$

Meanwhile, banks assume the riskier portion of the contract and must ensure they can be repaid under any reporting strategy. Therefore, banks impose their feasibility constraint, ensuring they are repaid independently of the firm's report:

$$\alpha A_t K_{t+1}^{\alpha-1} k_{jt} + (1 - \delta)k_{jt} \geq \max\{R_t^D (\omega_{jt} = 0 | \omega_{jt} = 1) L_{jt}^D, R_t^D (\omega_{jt} = 1) L_{jt}^D\} \tag{67}$$

This framework ensures that firms can pledge their full capital stock as collateral, but contract structure dictates claim priority:

- Bondholders' repayment is fully secured by the risk-free portion of capital since bondholders only accept risk-free debt, requiring full collateral backing.
- Banks accept risky debt, constrained by monitoring effectiveness and the firm's ability to repay under all reporting states. Thus, banks lend against the total productive resources of the firm, accepting risk in exchange for monitoring oversight.

Thus, while firms can pledge their full capital stock, the feasibility constraint ensures that bondholders' repayment is always secured first, and only the remaining risky portion is exposed to strategic misreporting and monitoring frictions. Finally, risk-free bonds and zero-recovery risky

bank loans can satisfy 65, 66 and 67. These conditions are the foundation of the financial contract structure analyzed in the main part of the paper.

E National accounts

A household's budget constraint is defined by:

$$c_{it}^a + d_{it+1}^a + b_{it+1}^a = (1 - \tau^g)p_H w_t z_{it} + (1 + r_t^d)d_{it} + (1 + r_t^b)b_{it} + \tau^f N_t^F(z_{it}) + \hat{r}d_{it}\mathbb{I}_{d_{it}<0}$$

$$c_{it}^n + d_{it+1}^n + b_{it} = (1 - \tau^g)p_H w_t z_{it} + (1 + r_t^d)d_{it} + (1 + r_t^b)b_{it} + \tau^f N_t^F(z_{it}) + \hat{r}d_{it}\mathbb{I}_{d_{it}<0}$$

Then, the average consumption and savings over a household's lifetime is:

$$\begin{aligned} & \nu_{it}c_{it}^a + (1 - \nu_{it})c_{it}^n + \nu_{it}d_{it+1}^a + (1 - \nu_{it})d_{it+1}^n + \nu_{it}b_{it+1}^a + (1 - \nu_{it})b_{it} \\ &= (1 - \tau^g)p_H w_t z_{it} + (1 + r_t^d)d_{it} + (1 + r_t^b)b_{it} + \tau^f N_t^F(z_{it}) + \hat{r}d_{it}\mathbb{I}_{d_{it}<0} \end{aligned}$$

where $w_t = A(1 - \alpha)K_t^\alpha$ and $\tau^f = \tau - \tau\zeta$ and $p_H w_t = W_t$. Then, in aggregate:

$$C_t^* + D_{t+1}^* + B_{t+1}^* = (1 - \tau^g)p_H A(1 - \alpha)K_t^\alpha + (1 + r_t^d)D_t^* + (1 + r_t^b)B_t^* + \tau^f N_t^F + \hat{r}D_t^- \quad (68)$$

where consumption is defined by $C_t^* = \int_i (\nu_{it}c_{it}^a + (1 - \nu_{it})c_{it}^n) d_i$, deposit holdings by $D_{t+1}^* = \int_i (\nu_{it}d_{it+1}^a + (1 - \nu_{it})d_{it+1}^n) d_i$, bond holdings by $B_t^* = \int_i (\nu_{it}b_{it+1}^a + (1 - \nu_{it})b_{it}) d_i$ and borrowing by $D^- = \int d_i \mathbb{I}_{d_{it}<0}$.

Entrepreneurs' total net worth is defined by:

$$N_t^E = (1 - \tau^f)N_t^F$$

Firms' net worth is defined by:

$$N_t^F = R_t^K K_t - R_t^B L_t^B - p_H R_t^D L_t^D$$

Therefore, in steady state $B_{t+1}^* = B_t^* \equiv B^*$ and $D_{t+1}^* = D_t^* \equiv D^*$

$$C^* + N^E = p_H A (1 - \alpha) K^\alpha - \tau^g p_H A (1 - \alpha) K^\alpha + r^d D^* + r^b B^* + N^F + \hat{r} D^-$$

where total taxes collected by government (T) satisfies the following: $T \equiv \tau^g p_H A (1 - \alpha) K^\alpha$ and $G = T$.

$$C^* + G^* + N^E = p_H A (1 - \alpha) K^\alpha + r^d D^* + r^b B^* + R^K K - R^B L^B - p_H R^D L^D + \hat{r} D^-$$

and after rearranging terms where $R^K = p_H \alpha A K^{\alpha-1} + p_H (1 - \delta) \kappa + (1 - \delta)(1 - \kappa)$, I obtain:

$$C^* + G^* + K^* (1 - p_H (1 - \delta) \kappa - (1 - \delta)(1 - \kappa)) = p_H A (1 - \alpha) K^{*\alpha} + p_H \alpha A K^{\alpha} - \hat{c} D^* + \hat{r} D^-$$

Finally, the resource constraint in steady state is defined by:

$$C^* + G^* + K^* (1 - (1 - \delta) (1 + p_H \kappa - \kappa)) + \hat{c} D^* - \hat{r} D^- = p_H A K^{*\alpha} \quad (69)$$

Therefore,

$$I = K^* (1 - (1 - \delta) (p_H \kappa + (1 - \kappa))) \quad (70)$$

F Numerical solution: Algorithm

Section 4 describes the model estimation. In this Appendix, I describe the details of the numerical implementation. First, I approximate the labor productivity process ($e_i \equiv \log z_i$) using a symmetric grid of n_e . Specifically, the grid points are:

$$e_i = \alpha \left(i - \frac{n_e - 1}{2} \right), i = 0, 1, 2, \dots, n_e \quad (71)$$

where the scaling parameter $\alpha = \frac{2\sigma_1}{\sqrt{n_e - 1}}$ is tied to the innovation standard deviation σ_1 in the initial steady state. Importantly, this grid is held fixed when computing the second stationary equilibrium as well. Moreover, I exponentiate the grid points to obtain $\tilde{z}_i = \exp e_i$. Note that

these $\tilde{z}_i$ are not normalized to have unit mean since normalization arises endogenously once the stationary distribution of the Markov chain is computed.

The transition probability matrix, $\mathbf{P}_z$, is approximated using a Monte Carlo method. For each state, I simulate shocks from the AR(1) process of log income and assign probabilities based on the simulated frequencies. Starting from a uniform distribution over states, the stationary distribution π is obtained iterating from $\pi^{k+1} = \pi^k \mathbf{P}_z$, until convergence. The resulting stationary distribution π is normalized to sum to one and used to determine the effective normalization of income levels.

Finally, the asset grid is log-spaced, with $n_d = n_b = 36$ grid points for deposits and bonds, respectively, resulting in a total of $n_d \times n_b \times n_z = 20,736$ grid points. To solve the household problem, I implement the endogenous grid method (EGM), following the approach of [Bayer et al. \(2019\)](#). This method provides an efficient way to handle borrowing constraints and portfolio choice, and the specific steps of the algorithm are outlined below.

F.1 Households

1. Define spaced grid nodes $\mathcal{H}_d = \{d_1, d_2, \dots, d_D\}$, $\mathcal{H}_b = \{b_1, b_2, \dots, b_B\}$, $\mathcal{H}_z = \{z_1, z_2, \dots, z_Z\}$ and transition matrix $\mathbf{P}$.³⁴ The state space is defined by $S = [i_{N_Z} \otimes b, i_{N_B} \otimes d, z \otimes i_{N_D}]$ where N_D, N_B and N_z are nodes for $\mathcal{H}_d, \mathcal{H}_b$ and $\mathcal{H}_z$. The number of nodes for S can be defined by $N_s = N_D \times N_B \times N_z$.

2. Define policy guesses:

$$c_n^{*(0)}(d, b, \mathbf{z}_{z \times 1}) = w\mathbf{z}\bar{l} + r^b b + (1 + r^d) d + \Pi$$

$$c_a^{*(0)}(d, b, \mathbf{z}_{z \times 1}) = w\mathbf{z}\bar{l} + (1 + r^b) b + (1 + r^d) d + \Pi$$

3. Start iteration with the policy functions $c_n^{*(0)}(d, b, z)$, $c_a^{*(0)}(d, b, z)$ and $\psi^{(0)}(d, b, z) = 0$ on a given grid $(d, b) \in D \times B$.
4. Update $c_n^*(d_n^*, b, z)$ in n^{th} iteration implementing EGM. In matrix form, for a given (d', b', z) (on-grid):

³⁴This matrix will be computed using Rouwenhorst algorithm to discretized a continuous Markov Chain of a AR(1) log-normal process.

$$\frac{\partial u(c_n^{*(n)}(d_n^*(d', b', z), b, z))}{\partial c} = \beta(1+r^d) \left(\nu^{*(n-1)} \mathbf{P}_z \frac{\partial u(c_a^{*(n-1)}(d', b', z'))}{\partial c} + (1-\nu^{*(n-1)}) \mathbf{P}_z \frac{\partial u(c_n^{*(n-1)}(d', b', z'))}{\partial c} \right) \quad (72)$$

Then, using approximant functions, interpolate $c_n^{*(n)}(d_n^*(d', b', z), b, \mathbf{z}_{z \times 1})$ at $c_n^{*(n)}(d, b, \mathbf{z}_{z \times 1})$. Regarding the binding constraints, I identify the points where the policy $d_n^* < -\underline{d}$ and adjust $d_n^* = -\underline{d}$ and the consumption policy accordingly.

- Using EGM evaluated at every $(b', z) \in \mathcal{H}_b \otimes \mathcal{H}_z$ (on-grid), I find $\tilde{d}_a^*(b', z)$ (off-grid) that solves the following condition:

$$\begin{aligned} 0 = & \beta \nu^{*(n-1)} \left((r^b - r^d) \mathbf{P}_z \frac{\partial u(c_a^{*(n-1)}(\tilde{d}_a^*(b', z), b', \mathbf{z}'_{z \times 1}))}{\partial c} \right) \\ & + \beta (1 - \nu^{*(n-1)}) \left((r^b - r^d) \mathbf{P}_z \frac{\partial u(c_n^{*(n-1)}(\tilde{d}_a^*(b', z), b', \mathbf{z}'_{z \times 1}))}{\partial c} \right) \\ & + \beta (1 - \nu^{*(n-1)}) \mathbf{P}_z \psi^{(n-1)}(\tilde{d}_a^*(b', z), b', \mathbf{z}'_{z \times 1}) \end{aligned} \quad (73)$$

If no solution exists, I set $\tilde{d}_a^*(b', z) = -\underline{d}$. Also, if solution exists, but if $\tilde{d}_a^*(b', z) < -\underline{d}$ then I set $\tilde{d}_a^*(b', z) = -\underline{d}$ and if $\tilde{d}_a^*(b', z) > d_D$, then I set $\tilde{d}_a^*(b', z) = d_D$.

- Solve for consumption for a given bond choice b' (on-grid) when there are bond adjustments $c_a^{*(n)}(b', z)$ using the following condition:

$$\frac{\partial u(c_a^{*(n)}(b', z))}{\partial c} = \beta(1+r^d) \left(\nu^{*(n-1)} \mathbf{P}_z \frac{\partial u(c_a^{*(n)}(\tilde{d}_a^*(b', z), b', z'))}{\partial c} + (1-\nu^{*(n-1)}) \mathbf{P}_z \frac{\partial u(c_n^{*(n)}(\tilde{d}_a^*(b', z), b', z'))}{\partial c} \right) \quad (74)$$

Next, find the total expenses $\mathcal{R}_a(b', z)$ which are consistent to total resources

$$\mathcal{R}_a(b', z) := c_a^{*(n)}(b', z) + d_a^{*(n)}(b', z) + b' - wz\bar{l} - \Pi$$

$$\tilde{\mathcal{R}}_a(d(b', z), b(b', z)) = (1 + r^d) d(b', z) + (1 + r^b) b(b', z)$$

Interpolating $c_a^{*(n)}(b', z')$, $d_a^{*(n)}(b', z')$ at $\tilde{\mathcal{R}}_a(d, b, z)$ that is consistent with $\tilde{\mathcal{R}}_a(d(b', z), b(b', z))$, I get policy functions $c_a^{*(n)}(d, b, z)$ and $d_a^{*(n)}(d, b, z)$.

7. Update ψ such that

$$\begin{aligned} \psi^{(n)}(d, b, z) = & \beta \nu^{*(n)} (1 + r^b) \mathbf{P}_z \frac{\partial u(c_a^{*(n)}(d_n^*(d, b, z), b, z'))}{\partial c} \\ & + \beta (1 - \nu^{*(n)}) (1 + r^b) \mathbf{P}_z \frac{\partial u(c_n^{*(n)}(d_n^*(d, b, z), b, z'))}{\partial c} \\ & + \beta (1 - \nu^{*(n)}) \mathbf{P}_z \psi^{(n-1)}(d_n^*(d, b, z), b, z') \end{aligned} \quad (75)$$

8. After convergence of policies, I update the value functions using the following conditions:

$$\begin{aligned} V_n^{(n)} &= u(c_n^{*(n)}) + \beta \mathbb{E}V(d_n^{*(n)}, b^{(n)}, z') \\ V_a^{(n)} &= u(c_a^{*(n)}) + \beta \mathbb{E}V(d_a^{*(n)}, b^{*(n)}, z') \\ \mathbb{E}V &= \nu^{*(n-1)} V_a^{(n)} + (1 - \nu^{*(n-1)}) V_n^{(n)} - \int_0^{V_a^{(n)} - V_n^{(n)}} \chi f(\chi) d\chi \end{aligned} \quad (76)$$

where the expected adjustment cost is defined by $AC(\nu; \mu_x, \sigma_x) = \int_0^{V_a^{(n)} - V_n^{(n)}} \chi f(\chi) d\chi = \mu_x \nu + \sigma_x (\nu \ln \nu + (1 - \nu) \ln (1 - \nu)) - C$.

9. Finally, update probabilities of adjustment $\nu^{*(n)} = F_\chi[V_a^{(n)} - V_n^{(n)}]$ since adjustment costs are distributed logistically.

10. Update probabilities of adjustment and iterate until convergence of policies.

F.2 Firms in the aggregate

1. Start with some guesses for interest rates, r^b and r^d and define the law of motion of aggregate stock of capital and firms' net worth where $L_t^B = \int_i b_{it} d\Theta_b$ and $L_t^D = \int_i d_{it} d\Theta_d$ represents the aggregate supply of bond savings and deposits collected by financial intermediaries in the stationary equilibrium.

$$K_{t+1}^n = N_t^n + L_t^B + L_t^D \quad (77)$$

$$N_{t+1}^n = (1 - \tau + \tau\zeta) \left(R_t^{*K} K_{t+1}^n - R_t^B (r_t^{*b}) L_t^B - p_H R_t^D (r_t^{*d}) L_t^D \right) \quad (78)$$

Iterate until convergence (i.e. $|N_{t+1}^n - N_{t+1}^{n-1}| < \epsilon$) and obtain stock of capital K_{SS}^* and net worth N_{SS}^* in the stationary equilibrium. By Corollary 1, to guarantee the convergence of net worth and positive returns to households (in equilibrium), the stock of assets must satisfy that $K_{SS}^* \in (\underline{K}, \bar{K})$.

2. Define the function of excess of demand of risk-free bonds and risky bank loans and find the interest rates that clears the market in four scenarios: i) both financial constraints are binding, ii) only risk-free bond demand is binding, iii) only risky bank loans is binding and iv) both financial constraint are slack.

$$L^B(K_{SS}^*, r^b, r^d) - B^*(r^b, r^d) = 0 \quad (79)$$

$$L^D(K_{SS}^*, r^b, r^d) - D^*(r^b, r^d) = 0 \quad (80)$$

and get the pair $\{r^b, r^d\}$ that solve the system of equations.

3. Find $\bar{R}^B$ and $\bar{R}^D$ and choose the equilibrium using the conditions in 35 and 37.
4. Find wages in equilibrium using 54 and compute the expected labor wage including the probability to have a positive wage:

$$w^{*e} = p_H(1 - \alpha)AK^{*\alpha} \quad (81)$$

5. Update interest rate and wages guesses and iterate households and firms algorithm until convergence.